



Design, Calculation, Proteus Scheme, Simulation, and Troubleshooting of a Current Limited LED Circuit Using a 9V Battery

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Report

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Abstract

This report presents the design and simulation of a basic current limited LED circuit powered by a 9V battery. The purpose of the project is to demonstrate why an LED cannot be safely connected directly to a voltage source without a current limiting resistor. The report explains the function of the resistor, the operating principle of the LED, and the reason current control is necessary in a practical electronic circuit. The circuit consists of a 9V battery, a red LED, a series resistor, connecting wires, and measuring instruments placed in the Proteus simulation environment. The theoretical foundation is based on the relationship between voltage, current, and resistance. The resistor is selected so that the current through the LED remains within a safe operating range. The report also introduces the role of Proteus ISIS Schematic Capture in constructing and testing the circuit before physical implementation. The expected outcome is a clear understanding of how a simple resistor protects an LED, how the circuit should be connected, and why incorrect resistor values can lead either to excessive current or to very weak LED brightness.

Keywords: Current limited LED circuit, 9V battery, red LED, resistor, Ohm's law, forward voltage, forward current, Proteus ISIS, circuit simulation, electronic protection, series circuit, voltage drop, current control, troubleshooting.

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Symbols and Abbreviations

This section defines the symbols and abbreviations used throughout the report. It is necessary because the project uses electrical quantities that must remain consistent across the calculation, scheme, simulation, and troubleshooting chapters.

(V_S) means supply voltage. In this project, the supply voltage is provided by the 9V battery.

(V_F) means LED forward voltage. It is the approximate voltage across the LED when it is forward biased and conducting current.

(V_R) means resistor voltage. It is the voltage dropped across the current limiting resistor.

(I_F) means LED forward current. It is the current flowing through the LED when the LED is operating in the forward direction.

(R) means resistance. It is measured in ohms.

(P_R) means resistor power dissipation. It is the electrical power converted into heat inside the resistor.

LED means light emitting diode.

DC means direct current.

ISIS means Intelligent Schematic Input System, the schematic capture environment used in Proteus.

1

Description

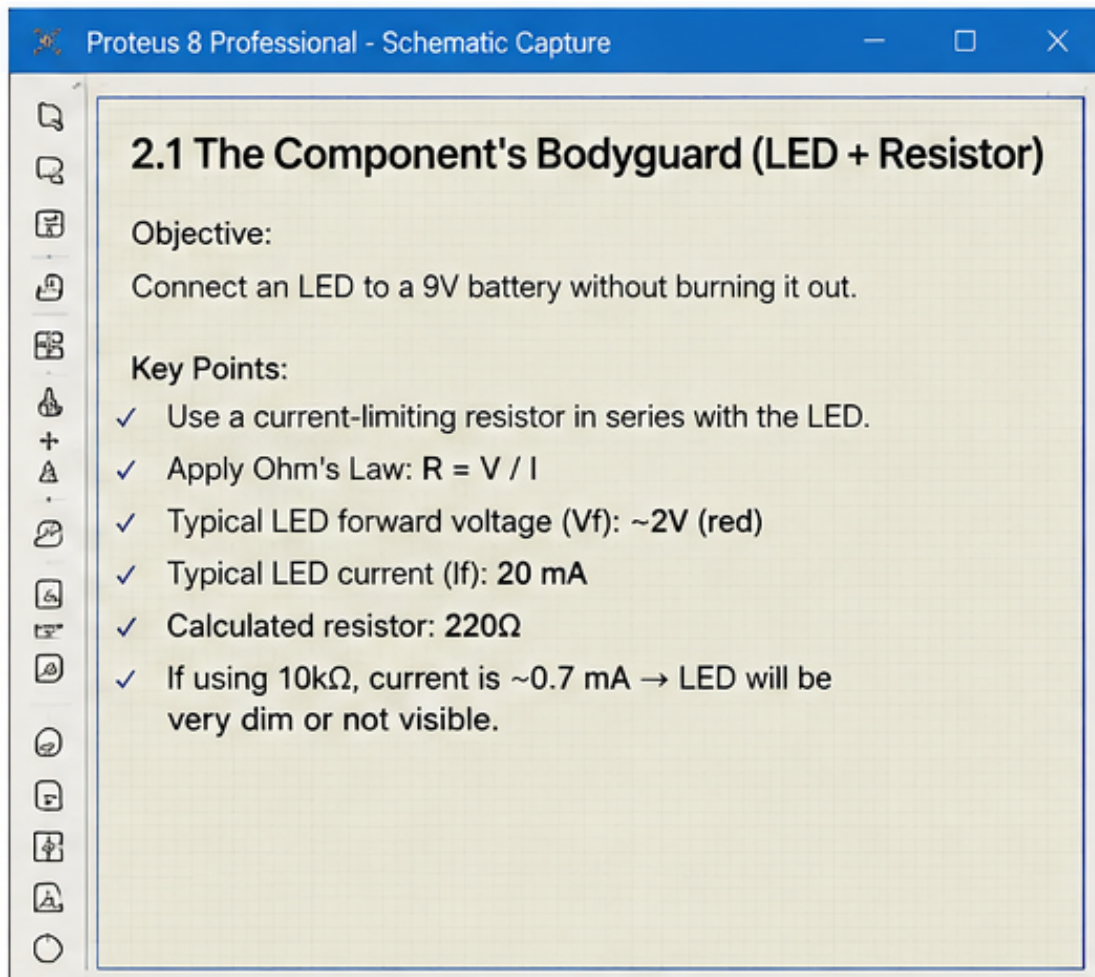


Figure 1.1: Description of the LED and resistor current limiting project

1.1 Introduction

This chapter introduces the design of a current limited LED circuit powered by a 9V battery. The project focuses on a fundamental principle of electronics: an LED must not be connected directly to a voltage source without a method of limiting current. Although an LED appears to be a simple component, it is sensitive to excessive current. If the current is not controlled, the LED may overheat, degrade, or fail permanently.

The circuit studied in this report uses a resistor in series with a red LED. The resistor controls the current by dropping the excess voltage that is not required by the LED. The design is implemented and tested in Proteus, which allows the circuit to be constructed, observed, and verified through simulation before any physical circuit is built.

The central idea of the project is that electronic components require controlled operating conditions. The LED is not protected by the battery itself. The battery supplies voltage, but the resistor establishes the safe current range. Therefore, the resistor acts as the protective component in the circuit.

1.2 Background of the Project

LEDs are widely used in electronic systems as indicators, display elements, signal devices, and visual output components. They are common in beginner circuits because they provide an immediate visible response when current flows through them. However, their simplicity can create a false impression that they can be connected directly to any battery. In reality, an LED requires correct polarity, suitable forward voltage, and safe forward current.

A red LED normally conducts when it is forward biased. This means the anode is connected toward the positive side of the supply and the cathode is connected toward the negative side. Once the LED begins conducting, its voltage remains relatively limited compared with the supply voltage. The remaining voltage must be controlled by another component. In this project, that component is a resistor.

The use of a resistor with an LED is one of the earliest and most important lessons in electronics. It teaches that voltage alone does not define safe operation. Current must also be controlled. The report therefore uses the LED circuit as a practical example of the relationship between theory and hardware protection.

1.3 Problem Statement

The problem addressed in this report is the unsafe connection of an LED to a 9V battery without proper current limitation. A standard red LED cannot safely absorb the full electrical effect of a 9V source. If the LED is connected directly to the battery, the current may rise beyond the safe operating value of the LED. This can cause excessive heating, brightness instability, internal damage, or complete failure.

The project also addresses a second problem: choosing an incorrect resistor value. If the resistor value is too small, the LED current may become too high. If the resistor value is too large, the LED current may become too low and the LED may appear very

dim or completely off. Therefore, the resistor value must be selected by calculation rather than guessing.

The report solves this problem by explaining the circuit, defining the components, introducing the required electrical relationships, and preparing the design for formal calculation and simulation.

1.4 Aim of the Project

The aim of this project is to design and explain a safe LED circuit powered by a 9V battery using a series current limiting resistor. The project aims to show how the resistor protects the LED by controlling the current flowing through the circuit.

The project also aims to demonstrate the correct use of Proteus for schematic construction and simulation. Through Proteus, the circuit can be tested visually and electrically. The LED state, voltage values, and current values can be observed before physical construction.

The wider aim is to connect theoretical electronics with practical circuit behavior. The report does not only show that the LED glows. It explains why the LED glows safely when the resistor value is suitable and why the LED behaves incorrectly when the resistor value is unsuitable.

1.5 Objectives of the Project

The first objective is to describe the function of each component in the LED circuit. The battery provides electrical energy, the resistor limits current, the LED produces light, the wires complete the electrical path, and the measuring instruments verify voltage and current.

The second objective is to explain why the resistor must be connected in series with the LED. In a series circuit, the same current flows through every component. This makes the resistor capable of directly controlling the LED current.

The third objective is to introduce the mathematical basis for resistor selection. The key relationship is that resistance is determined by voltage divided by current. In the LED circuit, the voltage across the resistor is the supply voltage minus the LED forward voltage.

The fourth objective is to prepare the circuit for Proteus implementation. The report must define the exact components and the correct connection order so that the circuit can be constructed without ambiguity.

The fifth objective is to prepare the reader for later simulation and troubleshooting. The reader should understand that normal LED operation, excessive current, weak brightness, and reversed polarity are all explainable by circuit theory.

1.6 Scope of the Project

The scope of this project is limited to a single LED circuit powered by a 9V DC source. The circuit uses one red LED and one series resistor. The project focuses on basic DC circuit behavior, current limitation, LED protection, and Proteus based simulation.

The project does not include complex LED driver circuits, pulse width modulation, microcontroller control, alternating current supply, thermal modeling, printed circuit board design, or advanced semiconductor physics. These topics are outside the intended level of the report.

The report concentrates on the essential design question: how can a red LED be connected to a 9V battery without damaging the LED? The answer is developed through component description, mathematical reasoning, schematic construction, simulation, and fault analysis.

1.7 Importance of Current Limiting in LED Circuits

Current limiting is essential because an LED is a current sensitive component. Unlike a resistor, an LED does not behave as a simple linear load. A small change in applied voltage can produce a large change in current once the LED is forward biased. For this reason, the current must be controlled by an external component.

The current limiting resistor protects the LED by taking the excess voltage from the supply. If the battery provides 9V and the LED requires approximately 2V in forward operation, the remaining voltage must appear across the resistor. The resistor then limits the current according to the relationship:

$$I = \frac{V_R}{R}$$

where (I) is the circuit current, (V_R) is the voltage across the resistor, and (R) is the resistance.

This relationship shows that the resistor value directly controls the LED current. A smaller resistor produces a larger current. A larger resistor produces a smaller current. Therefore, current limiting is not optional. It is the main protection mechanism in the circuit.

1.8 Basic Working Principle of an LED

An LED is a semiconductor device that emits light when current flows through it in the forward direction. It has two terminals: the anode and the cathode. The anode must be connected toward the positive side of the supply, and the cathode must be connected toward the negative side for normal operation.

When the LED is forward biased, it requires a certain forward voltage before significant current flows. For a red LED, this forward voltage is commonly treated as approximately 2V in basic circuit calculations. This value is not the full supply voltage. It is the approximate voltage used by the LED during conduction.

The LED brightness depends mainly on the forward current. If the current is too low, the LED may be dim or invisible. If the current is too high, the LED may become too bright for safe operation and may be damaged. Therefore, the LED must operate within a safe current range.

1.9 Why a Resistor Is Required

A resistor is required because the LED cannot safely regulate its own current when connected to a 9V battery. The battery supplies more voltage than the LED needs for forward conduction. The excess voltage must be dropped across another component, and the resistor performs this function.

The resistor value is selected using the relationship between voltage, current, and resistance. The resistor voltage is found by subtracting the LED forward voltage from the supply voltage:

$$V_R = V_S - V_F$$

The required resistance is then found by dividing the resistor voltage by the desired LED current:

$$R = \frac{V_S - V_F}{I_F}$$

In this project, the supply voltage is 9V. If the LED forward voltage is taken as approximately 2V and the desired LED current is 20mA, the resistor must control the remaining 7V. This prepares the calculation that will be completed in Chapter 2.

The resistor is therefore not an accessory. It is the component that makes the LED connection safe.

1.10 Overview of the 9V Battery LED Circuit

The circuit consists of a 9V battery connected in series with a resistor and a red LED. The current flows from the positive terminal of the battery through the resistor, then through the LED, and finally returns to the negative terminal of the battery.

The correct current path is:

Battery positive terminal to resistor input.

Resistor output to LED anode.

LED cathode to battery negative terminal.

Battery negative terminal to the circuit reference point when required in Proteus.

This arrangement creates a single closed loop. Since the resistor and LED are in series, the same current flows through both components. The resistor limits this current, while the LED converts part of the electrical energy into light.

The circuit is simple in structure but important in meaning. It demonstrates component protection, polarity, voltage distribution, current control, and simulation verification in one compact design.

1.11 Components Used in the Project

The circuit requires a small number of components, but each component has a specific role. The accuracy of the circuit depends on selecting the correct components and connecting them correctly. The main components are the 9V battery, the current limiting resistor, the red LED, connecting wires, and measuring instruments in Proteus.

The following component table should be included in the report:

Table 1.1: *Function of Each Circuit Component*

Component	Function in the Circuit
9V battery	Provides the DC supply voltage
Current limiting resistor	Controls the current through the LED
Red LED	Produces light when forward biased
Connecting wires	Complete the electrical path
Ammeter	Measures series current
Voltmeter	Measures voltage across selected components

Each component must be clearly labeled in the Proteus schematic so that the circuit can be understood without confusion.

1.11.1 9V Battery

The 9V battery is the voltage source of the circuit. It provides the electrical energy required for current to flow through the resistor and the LED. In the Proteus simulation, the battery should be set to 9V DC.

The positive terminal of the battery is connected to the resistor. The negative terminal is connected to the cathode side of the LED through the return path. This creates a complete circuit loop.

The 9V battery cannot be connected directly across the LED because the voltage is higher than the normal forward voltage of a red LED. Therefore, the battery must be used together with a current limiting resistor.

1.11.2 Current Limiting Resistor

The current limiting resistor is the protective component in the circuit. Its function is to restrict the current flowing through the LED. Without this resistor, the LED may draw excessive current from the 9V battery.

The resistor is connected in series with the LED. Since series components share the same current, the resistor directly determines the LED current. The resistor value must be calculated from the supply voltage, the LED forward voltage, and the desired LED current.

The general design formula is:

$$R = \frac{V_S - V_F}{I_F}$$

This formula shows that the resistor must absorb the voltage that remains after

the LED forward voltage is considered. The resistor also dissipates power as heat. Its power rating should therefore be suitable for the current and voltage conditions of the circuit.

1.11.3 Red LED

The red LED is the output component of the circuit. It emits light when forward current flows through it. The red LED has polarity, so it must be connected correctly. The anode must face the positive side of the circuit, and the cathode must face the negative side.

For basic calculation, the forward voltage of a red LED is commonly treated as approximately 2V. The desired forward current in this project is treated as approximately 20mA. These values allow the resistor value to be calculated in a clear and practical way.

If the LED is reversed, it will not conduct normally and will not glow. If the resistor value is too large, the LED may receive too little current and may appear dim. If the resistor value is too small, the LED may receive too much current and may be damaged.

1.11.4 Connecting Wires

Connecting wires provide the electrical path between the battery, resistor, LED, and measuring instruments. Although wires are simple components, incorrect wiring is one of the most common causes of circuit failure.

The wires must create a closed series loop. A broken wire causes an open circuit, which prevents current from flowing. A wrong wire connection may bypass the LED, reverse its polarity, or create an incorrect measurement path.

In the Proteus schematic, wires should be neat, direct, and easy to trace. Each wire should support the intended current path from the positive terminal of the battery through the resistor and LED back to the negative terminal.

1.11.5 Measuring Instruments in Proteus

Measuring instruments are used in Proteus to verify the circuit behavior. The ammeter measures current, and the voltmeter measures voltage. These instruments are essential because visual LED brightness alone is not sufficient evidence of correct circuit operation.

The ammeter must be connected in series with the circuit because it measures the current flowing through the loop. The voltmeter must be connected in parallel with the component whose voltage is being measured. For example, to measure LED voltage, the voltmeter is connected across the LED terminals.

The simulation should show both electrical values and visual behavior. A correct circuit should show a safe current value, an appropriate LED voltage, and normal LED brightness.

1.12 Software Used

The software used in this project is Proteus. Proteus allows the circuit to be constructed in a schematic environment and tested through simulation. It is suitable for this project because it supports electronic components, virtual instruments, DC sources, and visual circuit behavior.

The use of Proteus improves the reliability of the project because the circuit can be checked before physical construction. Errors in wiring, polarity, resistor value, and measurement placement can be identified and corrected in the simulation environment.

1.12.1 Proteus ISIS Schematic Capture

Proteus ISIS Schematic Capture is used to draw and organize the circuit diagram. It allows the user to place components, connect wires, assign component values, and prepare the circuit for simulation.

In this project, ISIS is used to place the 9V battery, current limiting resistor, red LED, ammeter, voltmeter, and ground reference when required. The schematic must be arranged clearly so that the current path can be followed from the battery positive terminal to the battery negative terminal.

A good schematic is not only a drawing. It is a technical representation of electrical relationships. Therefore, component labels, values, and connections must be accurate.

1.12.2 Simulation Environment

The simulation environment allows the circuit to operate virtually. When the simulation runs, Proteus calculates the circuit behavior and displays the state of the LED and the readings of the measuring instruments.

For this project, the simulation environment is used to verify that the resistor limits the current correctly. It is also used to test incorrect conditions, such as using a 10k Ω resistor or reversing the LED polarity.

The simulation must confirm the theoretical expectation. If the calculated current and simulated current are close, the design is verified. If the simulation result is very different from the calculation, the circuit must be checked for incorrect component values, wrong wiring, wrong polarity, or incorrect meter placement.

1.13 Expected Learning Outcomes

After completing this chapter and the full report, the reader should understand that an LED requires current limitation when connected to a voltage source. The reader should be able to explain why a resistor is connected in series with the LED and how the resistor protects the LED from excessive current.

The reader should also be able to identify the main components of the circuit and describe the function of each component. The reader should understand the current

path from the battery positive terminal through the resistor and LED back to the battery negative terminal.

The reader should be prepared to calculate the resistor value using the supply voltage, LED forward voltage, and desired LED current. The reader should also be prepared to construct the circuit in Proteus, run a simulation, measure voltage and current, and diagnose common circuit faults.

The most important learning outcome is that safe circuit design requires both correct theory and correct implementation. A circuit is not proven correct simply because it appears simple. It must be described, calculated, connected, simulated, and verified.

1.14 Chapter Summary

This chapter described the purpose, background, scope, components, and software environment of the current limited LED circuit. The main problem addressed in the chapter is that a red LED cannot be safely connected directly to a 9V battery without current limitation. The resistor is required because it controls the current flowing through the LED and protects it from excessive current.

The chapter explained that the LED must be connected with correct polarity and that the resistor must be placed in series with the LED. It also introduced the basic mathematical foundation for resistor selection, where the resistor value depends on the supply voltage, the LED forward voltage, and the desired LED current.

The chapter also described the role of Proteus ISIS Schematic Capture and the simulation environment. Proteus is used to construct the circuit, verify the current and voltage values, and observe the LED behavior before physical implementation.

Overall, Chapter 1 establishes the conceptual and technical foundation of the project. The next chapter should use this foundation to calculate the exact resistor value required for safe LED operation.

2

Calculation

2.1 Introduction to Circuit Calculation

The purpose of this chapter is to calculate the resistor value required to operate a red light emitting diode safely from a 9V battery. The calculation is necessary because an LED cannot be connected directly to a voltage source without current control. An LED is a polarity-sensitive semiconductor component, and once it becomes forward biased, its current can rise rapidly if no external limiting element is used. Therefore, a resistor must be placed in series with the LED to control the current and protect the component from excessive electrical stress.

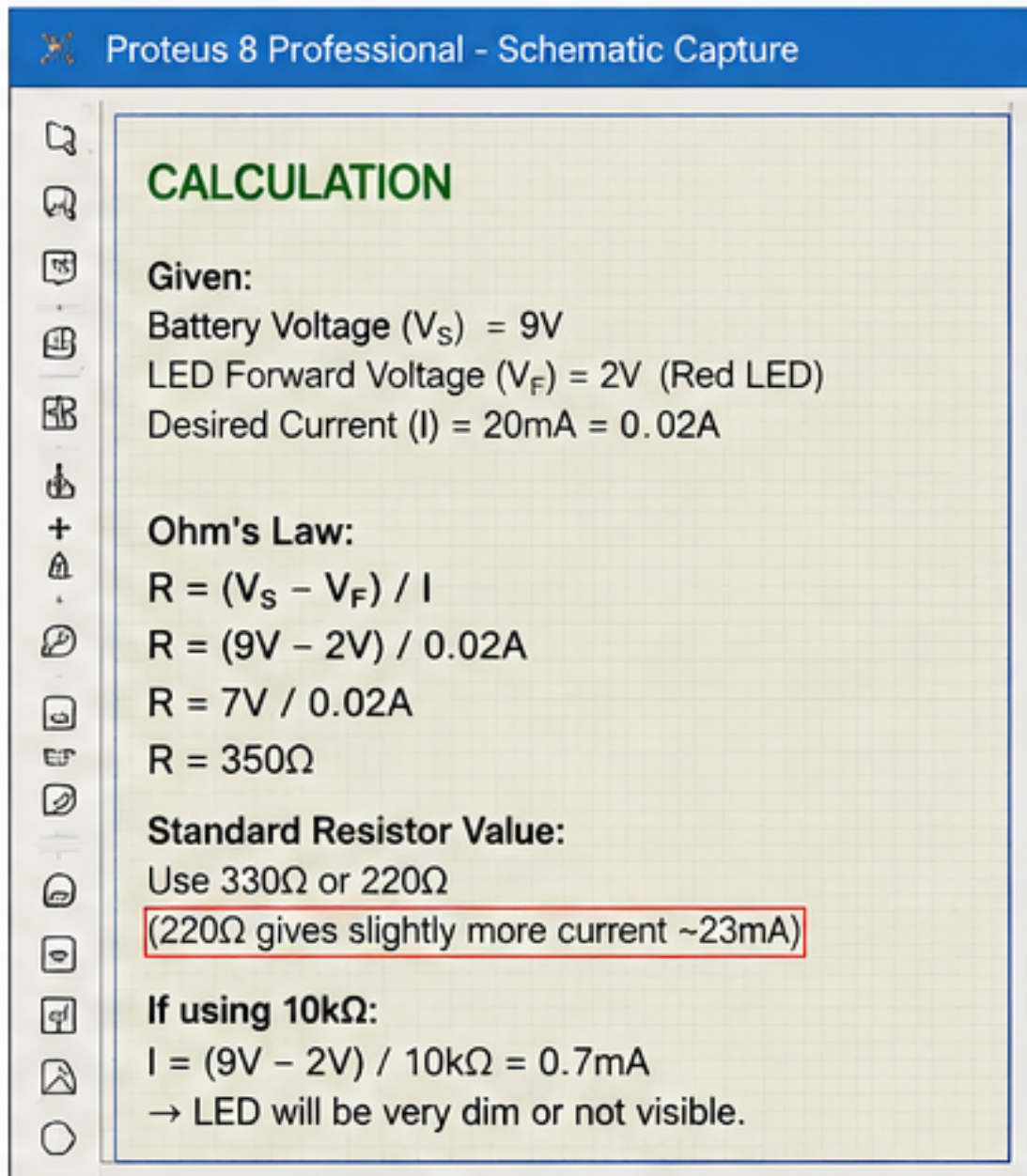
The calculation is based on the relationship between voltage, current, and resistance. In this project, the battery provides the total supply voltage, the LED requires a certain forward voltage to operate, and the remaining voltage must be dropped across the resistor. The resistor value is then selected so that the LED current remains within a safe and useful operating range.

This chapter establishes the exact numerical foundation of the circuit. It explains the given values, applies Ohm's law, determines the voltage across the resistor, calculates the ideal resistance, compares standard resistor values, evaluates current for different resistor choices, checks power dissipation, and finally selects the most appropriate resistor for the final circuit.

2.2 Given Circuit Parameters

The calculation begins by defining the electrical parameters of the circuit. These values determine the required resistor and must be clearly stated before any mathematical work is performed. The circuit consists of a 9V battery, a red LED, and a current-limiting resistor connected in series.

The assumed values for this project are as follows. The supply voltage is 9V. The forward voltage of the red LED is assumed to be 2V. The desired LED current is selected as 20mA, which is equal to 0.02A. These assumptions create a clear and consistent basis for the calculation.



The image shows a screenshot of the Proteus 8 Professional Schematic Capture window. The title bar at the top reads "Proteus 8 Professional - Schematic Capture". On the left side, there is a vertical toolbar with various icons for schematic capture. The main workspace has a light yellow background with a grid. The text in the workspace is as follows:

CALCULATION

Given:
Battery Voltage (V_S) = 9V
LED Forward Voltage (V_F) = 2V (Red LED)
Desired Current (I) = 20mA = 0.02A

Ohm's Law:
 $R = (V_S - V_F) / I$
 $R = (9V - 2V) / 0.02A$
 $R = 7V / 0.02A$
 $R = 350\Omega$

Standard Resistor Value:
Use 330 Ω or 220 Ω
(220 Ω gives slightly more current ~23mA)

If using 10k Ω :
 $I = (9V - 2V) / 10k\Omega = 0.7mA$
→ LED will be very dim or not visible.

Figure 2.1: Calculation of the current limiting resistor value

The calculation is valid under these stated assumptions. If a different LED type, forward voltage, or desired current is used, the resistor must be recalculated.

2.2.1 Supply Voltage

The supply voltage is the voltage provided by the battery. In this circuit, the source is a 9V battery. This means that the total electrical potential available to the series circuit is 9V.

The supply voltage is represented by the symbol (V_S).

$$V_S = 9V$$

This voltage is not applied entirely across the resistor because the LED also requires voltage when it is forward biased. Therefore, the supply voltage must be divided between the LED and the resistor.

2.2.2 LED Forward Voltage

The LED forward voltage is the voltage required across the LED for it to conduct current and emit light. For a typical red LED, the forward voltage is commonly approximated as 2V for basic design calculations.

The LED forward voltage is represented by the symbol (V_F).

$$V_F = 2V$$

This value means that approximately 2V of the 9V supply is used by the LED during normal operation. The remaining voltage must appear across the resistor.

The forward voltage is an approximation. In a real LED, the exact value depends on the LED model, current level, temperature, and manufacturing characteristics. For this academic circuit, 2V is used as the design assumption.

2.2.3 Desired LED Current

The desired LED current is the target current that should flow through the LED during normal operation. In this project, the selected current is 20mA.

The desired LED current is represented by the symbol (I_F).

$$I_F = 20mA$$

Since calculations using Ohm's law require amperes, the current is converted into amperes.

$$20mA = 0.02A$$

Therefore:

$$I_F = 0.02A$$

This current is selected because it is a common design target for a standard indicator LED. It is high enough to make the LED visible and low enough to remain within a typical safe operating range when the correct resistor is used.

2.3 Ohm's Law Foundation

The calculation is based on Ohm's law. Ohm's law states that voltage is equal to current multiplied by resistance.

$$V = I \times R$$

From this relationship, resistance can be calculated as voltage divided by current.

$$R = \frac{V}{I}$$

In this circuit, the resistor is responsible for dropping the voltage that remains after the LED forward voltage has been accounted for. Therefore, the resistor value is not calculated using the full 9V supply. Instead, it is calculated using the voltage across the resistor.

The correct form of the resistor calculation is:

$$R = \frac{V_S - V_F}{I_F}$$

This formula means that the LED forward voltage is subtracted from the supply voltage, and the remaining voltage is divided by the desired LED current.

2.4 Voltage Distribution in the Circuit

The circuit is a series circuit. In a series circuit, the same current flows through every component. Therefore, the resistor current and the LED current are equal.

The battery voltage is shared by the resistor and the LED. According to the voltage relationship in the loop:

$$V_S = V_R + V_F$$

Where (V_R) is the voltage across the resistor.

Since the supply voltage is 9V and the LED forward voltage is 2V, the resistor must drop the remaining voltage.

$$9V = V_R + 2V$$

Therefore:

$$V_R = 9V - 2V$$

$$V_R = 7V$$

This means that the resistor must safely drop 7V while allowing the desired current of 20mA to flow through the LED.

2.5 Voltage Across the Resistor

The voltage across the resistor is the part of the supply voltage that is not used by the LED. This voltage is essential because it is the voltage used in the resistor calculation.

The resistor voltage is:

$$V_R = V_S - V_F$$

Substituting the values:

$$V_R = 9V - 2V$$

$$V_R = 7V$$

Therefore, the resistor must be selected so that a 7V drop produces the desired current of 0.02A.

This is the central calculation point of the chapter. If the full 9V is incorrectly used in the resistor calculation, the selected resistance will be inaccurate because it ignores the voltage already dropped across the LED.

2.6 Resistor Value Calculation

The resistor value is calculated using Ohm's law.

$$R = \frac{V_R}{I_F}$$

The resistor voltage is 7V, and the desired current is 0.02A.

$$R = \frac{7V}{0.02A}$$

$$R = 350\Omega$$

Therefore, the ideal calculated resistor value is:

$$R = 350\Omega$$

This result means that a 350Ω resistor would allow approximately 20mA of current through the LED under the stated assumptions. Since 350Ω may not always be the most available standard resistor value, the final component selection must compare nearby standard values.

2.7 Standard Resistor Value Selection

The calculated value is 350Ω . In practical circuit construction, resistors are usually selected from standard manufactured values. Therefore, the ideal calculated value must be compared with nearby standard resistor values.

The nearest practical resistor values are commonly 330Ω , 360Ω , and 390Ω . Each value produces a different LED current. A lower resistor value produces higher current and higher brightness, while a higher resistor value produces lower current and lower brightness.

The standard resistor selection must balance brightness, safety, and accuracy relative to the desired 20mA current.

2.7.1 Calculated Value

The exact calculated value is:

$$R = 350\Omega$$

This value is obtained from:

$$R = \frac{9V - 2V}{0.02A}$$

$$R = 350\Omega$$

The calculated value represents the theoretical resistor required to produce exactly 20mA under the selected assumptions.

2.7.2 Nearest Standard Values

The nearest common standard values are:

$$330\Omega$$

$$360\Omega$$

$$390\Omega$$

The 330Ω resistor is slightly lower than the calculated value. Therefore, it allows slightly more than 20mA.

The 360Ω resistor is slightly higher than the calculated value. Therefore, it allows slightly less than 20mA and is very close to the design target.

The 390Ω resistor is higher than both 350Ω and 360Ω. Therefore, it produces a safer but slightly dimmer LED output.

2.7.3 Recommended Final Resistor Value

The recommended final resistor value for this project is 360Ω or 390Ω.

The 360Ω resistor is the strongest choice when the goal is to remain close to 20mA while still being slightly safer than the exact calculated value. The expected current with 360Ω is approximately 19.44mA.

The 390Ω resistor is the stronger conservative choice when safety and reduced current are prioritized. The expected current with 390Ω is approximately 17.95mA.

A 330Ω resistor may also function, but it produces approximately 21.21mA, which is slightly above the 20mA target. A 220Ω resistor should not be selected as the final academic design value for a strict 20mA target because it produces approximately 31.82mA.

2.8 Current Calculation for Different Resistor Values

The current through the LED can be calculated for different resistor values using the formula:

$$I = \frac{V_S - V_F}{R}$$

Since:

$$V_S - V_F = 7V$$

The current for each resistor is:

$$I = \frac{7V}{R}$$

This section compares five resistor values: 220Ω, 330Ω, 360Ω, 390Ω, and 10kΩ. The comparison shows how resistor selection directly controls LED current and brightness.

2.8.1 Current with 220Ω

For a 220Ω resistor:

$$I = \frac{7V}{220\Omega}$$

$$I = 0.031818A$$

Converting amperes to milliamperes:

$$I = 31.82mA$$

This current is significantly higher than the desired 20mA target. Although the LED may appear bright, this value is not the best safe academic design choice for a standard 20mA LED circuit. It increases electrical stress and may reduce LED lifetime or exceed the intended operating condition.

Therefore, 220Ω should be treated as an excessive-current case, not as the final recommended value.

2.8.2 Current with 330Ω

For a 330Ω resistor:

$$I = \frac{7V}{330\Omega}$$

$$I = 0.021212A$$

Converting amperes to milliamperes:

$$I = 21.21mA$$

This current is close to the 20mA target but slightly higher. In many simple demonstrations, this value may still produce a visible and functional LED circuit. However, from a strict calculation perspective, it is above the chosen design current.

Therefore, 330Ω is acceptable only if a small current increase is allowed. It is not the most precise protective choice.

2.8.3 Current with 360Ω

For a 360Ω resistor:

$$I = \frac{7V}{360\Omega}$$

$$I = 0.019444A$$

Converting amperes to milliamperes:

$$I = 19.44mA$$

This current is very close to the desired 20mA target and remains slightly below it. This makes 360Ω a strong practical choice for the final design.

The LED should glow clearly, and the current remains within the intended safe range.

2.8.4 Current with 390Ω

For a 390Ω resistor:

$$I = \frac{7V}{390\Omega}$$

$$I = 0.017949A$$

Converting amperes to milliamperes:

$$I = 17.95mA$$

This current is lower than the target current but still suitable for visible LED operation in most basic indicator applications. The LED may be slightly dimmer than it would be with 360Ω, but the circuit has a stronger safety margin.

Therefore, 390Ω is recommended when conservative current control is preferred.

2.8.5 Current with 10kΩ

For a 10kΩ resistor:

$$10k\Omega = 10000\Omega$$

$$I = \frac{7V}{10000\Omega}$$

$$I = 0.0007A$$

Converting amperes to milliamperes:

$$I = 0.7mA$$

This current is much lower than the desired 20mA target. At this current level, the LED may appear extremely dim or may not be visibly lit under normal room lighting.

Therefore, 10kΩ is not dangerous to the LED, but it is not suitable when normal LED brightness is required.

2.9 Resistor Power Dissipation Calculation

The resistor must also be checked for power dissipation. A resistor converts electrical energy into heat, and its power rating must be high enough to handle the expected power safely.

The resistor power can be calculated using:

$$P_R = V_R \times I$$

For the ideal calculated design:

$$V_R = 7V$$

$$I = 0.02A$$

Therefore:

$$P_R = 7V \times 0.02A$$

$$P_R = 0.14W$$

The resistor must therefore dissipate approximately 0.14W at the target current. A 0.25W resistor can normally handle this value, but a 0.5W resistor provides a stronger safety margin.

Power can also be calculated using:

$$P_R = I^2 \times R$$

For the calculated 350Ω resistor:

$$P_R = (0.02A)^2 \times 350\Omega$$

$$P_R = 0.0004 \times 350$$

$$P_R = 0.14W$$

Both methods give the same result. This confirms that the resistor should not be selected by resistance alone. Its power rating must also be suitable.

2.10 LED Safety Evaluation

The LED safety evaluation is based on current control. The LED is safest when the current remains at or below the desired operating current, provided the brightness is still acceptable.

The 220Ω resistor produces approximately 31.82mA. This is too high for a strict 20mA design and should not be used as the final recommended resistor.

The 330Ω resistor produces approximately 21.21mA. This is close to 20mA but slightly above the target. It may work, but it is not the most conservative choice.

The 360Ω resistor produces approximately 19.44mA. This is very close to the target and slightly below it. It is the strongest balanced choice.

The 390Ω resistor produces approximately 17.95mA. This gives lower brightness but stronger safety margin. It is the strongest conservative choice.

The 10k Ω resistor produces approximately 0.7mA. This is far below the target and makes the LED very dim or not visible.

Therefore, the safest and most academically justified resistor values are 360 Ω and 390 Ω . The final selection depends on whether the design prioritizes maximum closeness to 20mA or a stronger conservative margin.

2.11 Calculation Results Table

Table 2.1: Current Values and Power Results for Different Resistor Values

Resistor Value	Calculated Current	Resistor Power	LED Condition	Evaluation
220 Ω	31.82mA	0.223W	Very bright	Not recommended for a strict 20mA design
330 Ω	21.21mA	0.148W	Bright	Acceptable only with caution
350 Ω	20.00mA	0.140W	Normal brightness	Ideal calculated value
360 Ω	19.44mA	0.136W	Normal brightness	Strong recommended value
390 Ω	17.95mA	0.126W	Slightly dimmer but safe	Strong conservative value
10k Ω	0.70mA	0.0049W	Very dim or not visible	Troubleshooting case

The table shows that the resistor value directly determines the current through the LED. Lower resistance causes higher current, while higher resistance causes lower current. The ideal mathematical value is 350 Ω , but 360 Ω and 390 Ω are stronger practical choices because they keep the current below the 20mA design target.

2.12 Final Design Decision

The final design decision is to use a resistor value of 360 Ω or 390 Ω in series with the red LED.

If the purpose is to remain very close to the 20mA design target, 360 Ω should be selected. It gives approximately 19.44mA, which is almost equal to the desired current while remaining slightly below it.

If the purpose is to maximize safety margin and reduce stress on the LED, 390 Ω should be selected. It gives approximately 17.95mA, which is lower than the target but still suitable for visible LED operation.

The 220 Ω resistor should not be selected for the final circuit because it gives approximately 31.82mA, which is excessive for a strict 20mA design. The 10k Ω resistor should not be selected for normal operation because it gives only approximately 0.7mA, which makes the LED very dim or invisible.

Therefore, the final recommended design is:

$$V_S = 9V$$

$$V_F = 2V$$

$$I_F = 20mA$$

$$R_{calculated} = 350\Omega$$

$$R_{recommended} = 360\Omega$$

or:

$$R_{recommended} = 390\Omega$$

The final circuit must place this resistor in series with the LED so that the resistor controls the current through the entire circuit.

2.13 Chapter Summary

This chapter calculated the current-limiting resistor required for a red LED powered by a 9V battery. The LED forward voltage was assumed to be 2V, and the desired LED current was selected as 20mA. The voltage across the resistor was calculated as 7V, because the LED uses approximately 2V of the 9V supply.

Using Ohm's law, the ideal resistor value was calculated as 350Ω. Since practical circuits normally use standard resistor values, 330Ω, 360Ω, and 390Ω were compared. The 360Ω resistor produced approximately 19.44mA, while the 390Ω resistor produced approximately 17.95mA. Both values are suitable and safe choices. The 220Ω resistor produced approximately 31.82mA, which is too high for a strict 20mA design. The 10kΩ resistor produced approximately 0.7mA, which is too low for normal LED brightness.

The chapter also calculated resistor power dissipation and showed that the resistor should be rated safely above the expected heat dissipation. The strongest final design choice is therefore a 360Ω resistor for close target performance or a 390Ω resistor for a more conservative safety margin.

3

Scheme

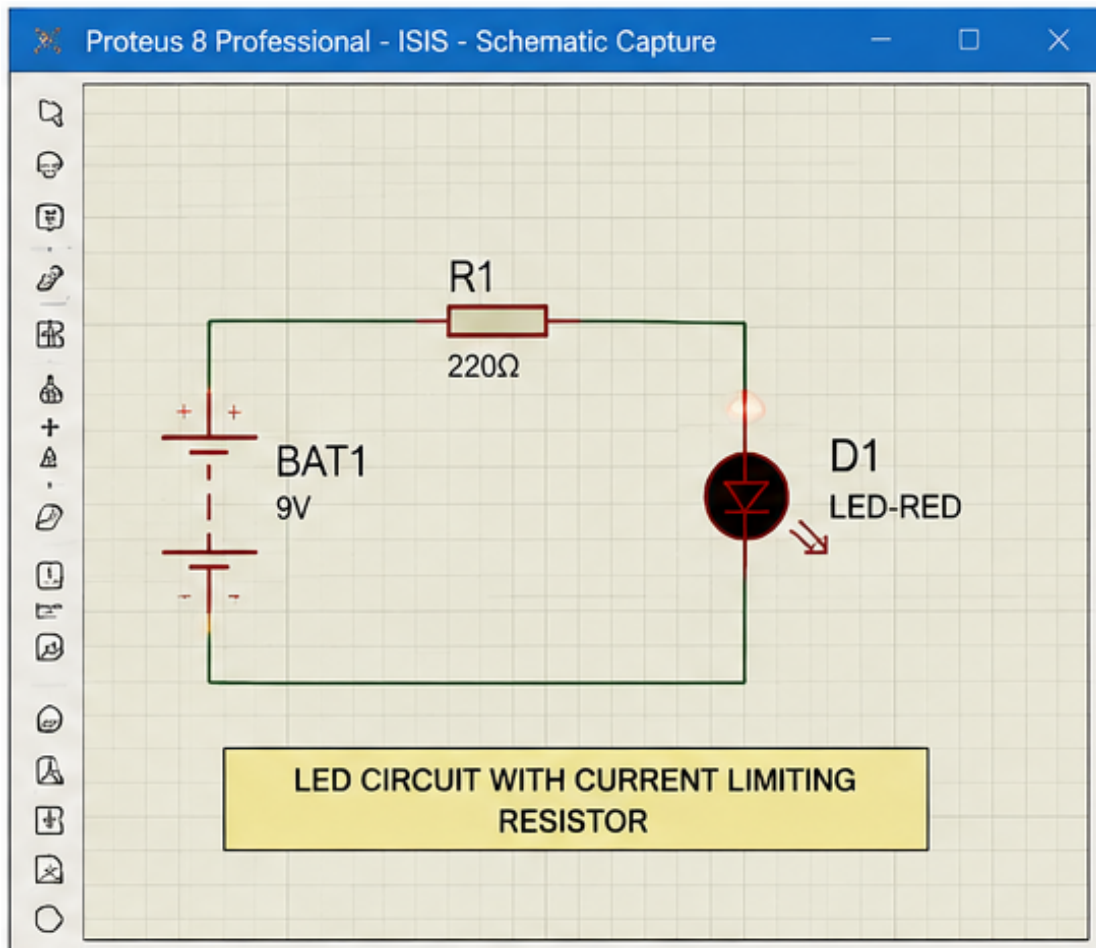


Figure 3.1: Proteus schematic of the LED circuit

3.1 Introduction to the Circuit Scheme

The circuit scheme represents the physical and electrical translation of the theoretical LED protection design into a structured Proteus schematic. The purpose of this chap-

ter is to define the exact components, the exact connection order, the correct polarity arrangement, the measurement positions, and the final schematic layout required for a safe current limited LED circuit powered by a 9V battery.

In this project, the scheme is not only a visual drawing of the circuit. It is the controlled electrical structure that proves how the battery, resistor, LED, ammeter, voltmeter, and ground terminal are arranged to form one valid operating circuit. A correct scheme must show that the LED is not connected directly to the battery. It must also show that the resistor is placed in series with the LED so that the current passing through the LED is limited to a safe value.

The circuit scheme must therefore satisfy three academic requirements. First, it must match the calculation from the previous chapter. Second, it must be physically reproducible in Proteus. Third, it must allow measurement of current and voltage during simulation. If any of these requirements is missing, the scheme becomes incomplete because it would not fully connect the theoretical calculation to the practical simulation.

For the final recommended design, the circuit uses a 9V battery, one current limiting resistor, and one red LED. The recommended resistor value is $360\ \Omega$ when the target current is close to 20 mA. A $390\ \Omega$ resistor may also be used when a more conservative current value is desired. The resistor is placed in series with the LED, meaning that the same current flows through the battery, resistor, and LED.

The basic electrical path of the circuit is as follows:

Battery positive terminal $\rightarrow$ Resistor $\rightarrow$
LED anode $\rightarrow$ LED cathode $\rightarrow$
Battery negative terminal

This path must remain unbroken. If the path is interrupted, no current flows. If the LED is reversed, it does not conduct in the intended direction. If the resistor is removed, the LED is exposed to excessive current. Therefore, the scheme chapter has the responsibility of proving that the circuit is electrically valid before simulation begins.

3.2 Proteus Component Selection

The Proteus schematic must be created with components that correctly represent the real circuit elements used in the project. Component selection is important because an incorrect component model can produce inaccurate simulation results. The selected components must correspond to the intended circuit function, the intended voltage level, and the intended measurement purpose.

The required Proteus components are the battery component, the resistor component, the red LED component, the ammeter component, the voltmeter component, and the ground terminal. Each component has a specific function in the circuit, and each one must be placed with correct orientation and correct electrical connection.

The battery provides the electrical energy. The resistor limits current. The LED

emits light when forward biased. The ammeter measures the current flowing through the series path. The voltmeter measures the voltage across a selected component. The ground terminal provides a reference node for the simulation environment when required.

The scheme must not include unnecessary components that do not serve the project aim. A clean academic schematic should include only the elements needed to demonstrate current limitation, LED operation, and electrical measurement.

3.2.1 Battery Component

The battery component represents the 9V direct current supply used to power the circuit. In Proteus, the battery must be configured so that its voltage is set to 9V. This value is essential because the entire calculation depends on the difference between the supply voltage and the LED forward voltage.

The positive terminal of the battery is the starting point of the conventional current path. The negative terminal of the battery is the return point of the circuit. The scheme must clearly show these two terminals because incorrect battery polarity would prevent the intended circuit operation.

The battery positive terminal must be connected to the input side of the current limiting resistor. The battery negative terminal must be connected to the LED cathode through the return wire. If a ground terminal is required by the Proteus simulation environment, the battery negative terminal must also be connected to ground.

The battery must not be connected directly across the LED. A direct connection would remove the current limiting function of the resistor and would create an unsafe design. The battery must always feed the LED through the series resistor.

3.2.2 Resistor Component

The resistor component is the protective element of the circuit. Its purpose is to limit the current flowing through the LED by dropping the excess voltage that remains after the LED forward voltage is accounted for.

For the recommended design, the resistor should be set to 360 Ω . This value gives a current close to 20 mA when the LED forward voltage is assumed to be approximately 2V. The expected current is calculated as:

$$I = \frac{9V - 2V}{360\Omega}$$

$$I \approx 0.0194A$$

$$I \approx 19.4mA$$

This current is close to the common target current for a standard red LED. A 390 Ω resistor may also be used if the designer wants a slightly lower and safer current. With 390 Ω , the expected current is:

$$I = \frac{9V - 2V}{390\Omega}$$

$$I \approx 0.0179A$$

$$I \approx 17.9mA$$

The resistor must be placed in series with the LED. It may be placed before the LED or after the LED in the same loop, but for this project it should be placed between the battery positive terminal and the LED anode. This position makes the circuit easier to read and makes the protective role of the resistor clear.

The resistor must be labeled as R1. Its value must be visible in the Proteus schematic. The label should show either 360 Ω or 390 Ω depending on the selected final design value.

3.2.3 LED Component

The LED component represents the red light emitting diode used as the output device of the circuit. In Proteus, the selected component should be a red LED model. The LED must be connected with correct polarity because it is a directional semiconductor device.

The LED has two terminals. The anode is the positive side, and the cathode is the negative side. For the LED to conduct and emit light, the anode must be connected toward the positive side of the circuit through the resistor, and the cathode must be connected toward the battery negative terminal.

The LED forward voltage is assumed to be approximately 2V for the calculation. This assumed voltage is not the full supply voltage. The remaining voltage is dropped across the resistor. Therefore, the LED and resistor share the total battery voltage according to their electrical behavior.

The LED must be labeled as D1 in the schematic. The component name should also identify it as a red LED. This labeling makes the schematic readable and allows the reader to understand the circuit without external explanation.

3.2.4 Ammeter Component

The ammeter component is used to measure the current flowing through the circuit. Since the LED and resistor are connected in series, the same current flows through both components. Therefore, the ammeter must also be connected in series with the circuit.

The ammeter must not be connected in parallel with the LED, resistor, or battery. A parallel ammeter connection would be incorrect because an ideal ammeter has very low internal resistance. Placing it in parallel can create an unintended current path and can distort or damage the simulated circuit behavior.

The recommended position for the ammeter is between the resistor and the LED anode, or between the battery positive terminal and the resistor. Either position is

valid because the circuit current is the same at every point in the series path. For clarity, placing the ammeter between R1 and D1 is useful because it directly shows the current entering the LED.

The ammeter reading should be close to the theoretical current. With a $360\ \Omega$ resistor, the expected reading is approximately 19.4 mA, assuming the LED forward voltage is approximately 2V. With a $390\ \Omega$ resistor, the expected reading is approximately 17.9 mA.

3.2.5 Voltmeter Component

The voltmeter component is used to measure the voltage across a selected circuit element. Unlike an ammeter, a voltmeter must be connected in parallel with the component being measured.

For this circuit, the primary voltmeter should be connected across the LED. The positive terminal of the voltmeter must connect to the LED anode, and the negative terminal of the voltmeter must connect to the LED cathode. This connection allows the simulation to display the approximate forward voltage of the LED.

A second voltmeter may be connected across the resistor if the report requires confirmation of resistor voltage. The positive terminal of this voltmeter should connect to the resistor input side, and the negative terminal should connect to the resistor output side. The expected voltage across the resistor is approximately:

$$V_R = 9V - 2V$$

$$V_R = 7V$$

The voltmeter must never be placed in series with the circuit because a voltmeter has very high internal resistance. A series voltmeter would block or severely reduce current and would make the LED fail to operate correctly.

3.2.6 Ground Terminal

The ground terminal provides the zero volt reference point for the circuit simulation. In many Proteus circuits, a ground reference is required to allow the simulator to define voltage levels correctly. For this project, the ground terminal should be connected to the battery negative terminal.

The ground terminal does not mean that the circuit is physically connected to earth in the real world. In the schematic, it functions as the reference node from which other voltages are measured. It defines the battery negative terminal as 0V.

The ground terminal should be labeled clearly and placed near the battery negative terminal. The connection must be direct and unambiguous. If the ground terminal is missing and Proteus requires one, the simulation may not run correctly or may produce unstable results.

3.3 Circuit Connection Method

The circuit must be connected as a single closed series loop. A closed loop means that there is a continuous path from the positive terminal of the battery, through the resistor, through the LED, and back to the negative terminal of the battery.

The resistor and LED must not be placed in parallel for this project. A parallel arrangement would not force the resistor to limit the LED current in the intended way. The protective behavior depends on the resistor being in the same current path as the LED.

The connection method must follow the principle that all current entering the LED first passes through the resistor. This ensures that the resistor controls the current before it reaches the LED. The circuit must therefore be arranged so that no wire bypasses the resistor.

The correct current path is:

$$+9V \rightarrow R1 \rightarrow D1 \rightarrow 0V$$

In this path, R1 represents the current limiting resistor and D1 represents the red LED. The total battery voltage is distributed between these two elements. The LED receives approximately its forward voltage, and the resistor receives the remaining voltage.

The schematic must show direct wire connections between component terminals. Wires must meet at clear connection points. Junctions must be visually clear so that the reader can distinguish connected wires from crossing wires.

3.4 Exact Wire Connection Order

The wiring sequence must be presented in a precise order so that the circuit can be reproduced without interpretation. The order begins at the battery positive terminal and ends at the battery negative terminal.

The complete wiring sequence is:

First, connect the positive terminal of the 9V battery to the first terminal of resistor R1. Second, connect the second terminal of resistor R1 to the anode of LED D1. Third, connect the cathode of LED D1 to the negative terminal of the 9V battery. Fourth, connect the battery negative terminal to the ground terminal if a simulation reference is required. Fifth, insert the ammeter in series with the main current path. Sixth, connect the voltmeter in parallel with the LED.

This order ensures that the resistor, LED, and ammeter all share the same current path, while the voltmeter measures voltage without interrupting the circuit.

3.4.1 Battery Positive to Resistor

The first wire connection must join the positive terminal of the battery to the input terminal of the resistor. This connection supplies electrical potential from the 9V source

to the current limiting element.

This wire is the beginning of the active current path. It must not branch directly to the LED. If the battery positive terminal is connected directly to the LED anode while bypassing the resistor, the resistor will not protect the LED.

The schematic should show the battery on the left side and the resistor on the upper part of the circuit. This layout makes the flow of the circuit easy to read from left to right.

3.4.2 Resistor to LED Anode

The second wire connection must join the output terminal of the resistor to the anode of the LED. This connection ensures that all current reaching the LED has already passed through the resistor.

The anode side of the LED must be identified correctly before the wire is connected. In Proteus, the LED symbol must be checked carefully to confirm which terminal is the anode and which terminal is the cathode.

If an ammeter is included between the resistor and LED, the connection becomes two wire segments. The resistor output connects to the ammeter input, and the ammeter output connects to the LED anode. This still counts as the same series path because the ammeter is inserted into the current path rather than placed across a component.

3.4.3 LED Cathode to Battery Negative

The third wire connection must join the cathode of the LED to the negative terminal of the battery. This connection completes the return path of the circuit.

The cathode must not be connected to the battery positive terminal in the final operating circuit. If the LED is reversed, the circuit becomes reverse biased for the LED, and the LED does not emit light under normal conditions.

This connection closes the circuit loop. Without this return wire, the circuit is open and current cannot flow. An open return path would produce a zero current reading on the ammeter and no light from the LED.

3.4.4 Battery Negative to Ground

The fourth connection links the battery negative terminal to the ground terminal. This connection defines the battery negative terminal as the 0V reference node for simulation.

The ground connection should be short, clear, and placed near the battery. It should not be drawn in a way that makes it appear to bypass the LED or resistor. The ground terminal is only a reference point and must not alter the intended current path.

When this connection is used, the node containing the battery negative terminal, LED cathode, and ground terminal becomes the ground node. This node is the reference point for voltage measurements.

3.4.5 Ammeter Series Connection

The ammeter must be inserted into the main series path. A correct position is between the resistor and the LED anode. Another valid position is between the battery positive terminal and the resistor. The selected position should be shown clearly in the schematic.

If the ammeter is placed between the resistor and LED, the connection order becomes:

Battery positive → Resistor →
Ammeter → LED anode

The ammeter must measure the same current that flows through the LED. This current is the most important measurement for confirming that the resistor is protecting the LED.

The expected ammeter reading for a 360 Ω resistor is approximately 19.4 mA under the assumed LED forward voltage. The expected reading for a 390 Ω resistor is approximately 17.9 mA. If the ammeter shows a value much higher or much lower than expected, the circuit should be checked for incorrect resistance, incorrect LED polarity, broken wires, or incorrect meter placement.

3.4.6 Voltmeter Parallel Connection

The voltmeter must be connected across the LED to measure the LED forward voltage. The positive terminal of the voltmeter connects to the LED anode. The negative terminal of the voltmeter connects to the LED cathode.

This connection allows the voltmeter to measure the potential difference across the LED without interrupting the main current path. The expected LED voltage is approximately 2V for a red LED, although the exact value can vary depending on the LED model used in Proteus.

If a second voltmeter is used across the resistor, it should measure approximately 7V when the LED forward voltage is approximately 2V and the supply voltage is 9V. The voltage relationship is:

$$V_S = V_R + V_F$$

$$9V \approx 7V + 2V$$

This relationship confirms that the supply voltage is distributed between the resistor and the LED.

3.5 LED Polarity Identification

Correct LED polarity is essential because an LED conducts primarily in one direction. The anode must be connected toward the positive side of the circuit, and the cathode

must be connected toward the negative side of the circuit.

In the Proteus schematic, the LED symbol must be inspected before final wiring. The anode is the terminal that receives current after it passes through the resistor. The cathode is the terminal that returns current to the battery negative terminal.

If the LED polarity is correct, the LED becomes forward biased. In this condition, current can pass through the LED, and the LED emits light. If the polarity is reversed, the LED becomes reverse biased. In that condition, the LED does not operate as intended, and the simulation will show little or no forward current.

The polarity rule for this circuit is:

Resistor output $\rightarrow$ LED anode

LED cathode $\rightarrow$ Battery negative

This polarity arrangement must be preserved in the final schematic and in all simulation images.

3.6 Series Circuit Arrangement

The circuit is a series arrangement because the battery, resistor, LED, and ammeter are connected in one continuous path. In a series circuit, the same current flows through every component in the path.

The series arrangement is necessary because the resistor must control the same current that passes through the LED. If the resistor is not in series, it cannot reliably limit the LED current.

The current relationship in the series circuit is:

$$I_{\text{battery}} = I_R = I_{\text{LED}}$$

This means that the current supplied by the battery is equal to the current through the resistor and equal to the current through the LED. This equality is the central reason why a series resistor protects the LED.

The voltage relationship in the circuit is:

$$V_S = V_R + V_{\text{LED}}$$

For the assumed values:

$$9V \approx 7V + 2V$$

The resistor absorbs the voltage that is not used by the LED. This prevents excessive current from passing through the LED.

The final scheme must therefore show one uninterrupted loop. Any additional wire that bypasses the resistor or bypasses the LED would destroy the intended series

structure and would make the schematic invalid.

3.7 Node Identification

Node identification is used to describe the electrical points in the schematic where component terminals are connected. A node is a continuous conductor region with the same voltage. Correct node identification makes the circuit easier to analyze and verify.

This circuit has three main nodes. The first node is the positive supply node. The second node is the resistor to LED junction node. The third node is the ground or negative node.

These nodes are sufficient to describe the complete circuit. They also allow the reader to understand where voltage measurements should be taken and where current flows.

3.7.1 Positive Supply Node

The positive supply node is the point connected to the positive terminal of the 9V battery. This node is the highest potential point in the circuit relative to ground.

The positive supply node connects the battery positive terminal to the input side of resistor R1. If the ammeter is placed before the resistor, this node may also connect to one terminal of the ammeter. The exact placement depends on the selected measurement layout.

This node should be labeled as +9V or Battery Positive. The label must be clear because it identifies the starting point of the current path.

3.7.2 Resistor LED Junction Node

The resistor LED junction node is the point between the resistor output and the LED anode. This node is important because it represents the voltage after the resistor has dropped part of the supply voltage.

If the LED forward voltage is approximately 2V and the LED cathode is connected to ground, the resistor LED junction node should be near 2V relative to ground. This is because the LED anode sits at approximately the LED forward voltage when the LED is conducting.

This node may also connect to the positive terminal of the voltmeter used to measure LED voltage. If the ammeter is placed between the resistor and LED, the node may be divided visually by the ammeter, but electrically it remains part of the same series path.

The node should be labeled as LED Anode Node or Resistor Output Node. The label must not imply that this node is the full 9V supply, because the resistor has already caused a voltage drop.

3.7.3 Ground Negative Node

The ground negative node is the return node of the circuit. It connects the LED cathode, the battery negative terminal, and the ground terminal.

This node is normally defined as 0V in the simulation. It is the reference point for measuring LED voltage, resistor voltage, and supply voltage.

The voltmeter negative terminal should connect to this node when measuring LED voltage. The battery negative terminal must also connect to this node to close the circuit.

This node should be labeled as 0V, Ground, or Battery Negative. The label must be consistent throughout the report.

3.8 Circuit Labeling Requirements

The schematic must include clear labels for every essential component and value. Labeling is necessary because the circuit must be understandable without verbal explanation.

The battery should be labeled BAT1 and its value should be shown as 9V. The resistor should be labeled R1 and its value should be shown as 360 Ω or 390 Ω . The LED should be labeled D1 and identified as a red LED. The ammeter should be labeled as the current measuring instrument. The voltmeter should be labeled as the voltage measuring instrument.

The labels should be placed close to the components but must not overlap wires or symbols. A clean schematic is academically stronger because it reduces interpretation errors.

The current direction may be indicated with a small arrow if the report format allows it. The arrow should follow the path from battery positive to resistor, from resistor to LED anode, from LED cathode to battery negative.

The final schematic should also include a figure title, such as:

Figure 3.1: Proteus schematic of the 9V LED circuit with a series current limiting resistor.

The figure title should be placed below the image in the academic report. The title must describe the function of the schematic, not only name the image.

3.9 Final Proteus Scheme Layout

The final Proteus scheme should be arranged in a clean rectangular or loop based layout. The battery should be positioned on the left side. The resistor should be positioned on the upper path. The LED should be positioned on the right side or lower right side. The return wire should connect the LED cathode back to the battery negative terminal.

The layout should allow the current path to be read naturally. The reader should be able to follow the circuit from the battery positive terminal, through R1, through D1, and back to the battery negative terminal without confusion.

The recommended final layout is:

Left side: BAT1

Top path: R1

Right side: D1

Bottom path: return wire to BAT1 negative

The ammeter should be placed in the same series path. The voltmeter should be placed across the LED, with its terminals connected to the LED anode and cathode.

The final scheme must not include disconnected wires, floating terminals, reversed measuring instruments, or unclear junctions. Every wire must terminate at a valid component terminal or node.

The final scheme should visually prove the following facts. The resistor is in series with the LED. The LED is forward biased. The battery value is 9V. The resistor value is visible. The ammeter is in series. The voltmeter is in parallel with the LED. The ground node is connected to the battery negative terminal if required.

3.10 Scheme Verification Checklist

The scheme must be verified before running the simulation. Verification ensures that the schematic is not only visually correct but electrically valid.

The battery must be checked first. Its voltage must be set to 9V, and its positive and negative terminals must be correctly identified.

The resistor must be checked second. Its value must match the selected design value. If the project uses the recommended near 20 mA design, R1 should be 360 Ω . If the project uses a more conservative design, R1 should be 390 Ω .

The LED must be checked third. Its anode must connect toward the resistor and battery positive side. Its cathode must connect toward the battery negative side.

The ammeter must be checked fourth. It must be inserted in series with the circuit. It must not be placed across the battery, resistor, or LED.

The voltmeter must be checked fifth. It must be connected in parallel with the LED if the LED voltage is being measured. Its positive terminal must connect to the LED anode, and its negative terminal must connect to the LED cathode.

The ground terminal must be checked sixth. It should be connected to the battery negative terminal if required by the simulation.

The full circuit must be checked seventh. The circuit must form a closed loop from battery positive to battery negative. There must be no broken wires, no floating terminals, and no unintended bypass paths.

The mathematical expectation must be checked last. With a 360 Ω resistor and an

assumed 2V LED forward voltage, the current should be approximately 19.4 mA. With a 390 Ω resistor, the current should be approximately 17.9 mA. These values prepare the report for the simulation chapter.

The verification logic is:

$$I = \frac{V_S - V_F}{R}$$

For 360 Ω :

$$I = \frac{9V - 2V}{360\Omega}$$

$$I \approx 19.4mA$$

For 390 Ω :

$$I = \frac{9V - 2V}{390\Omega}$$

$$I \approx 17.9mA$$

If the simulated current is far from these values, the scheme must be inspected again before accepting the simulation result.

3.11 Chapter Summary

This chapter presented the complete circuit scheme for the 9V LED protection project. The scheme uses a battery, a current limiting resistor, a red LED, an ammeter, a voltmeter, and a ground reference. The chapter explained the function of each component, the correct wiring order, the required LED polarity, the series circuit arrangement, the main circuit nodes, the labeling requirements, the final Proteus layout, and the verification checklist.

The most important requirement of the scheme is that the resistor must be connected in series with the LED. This arrangement ensures that the same current flows through both the resistor and the LED, allowing the resistor to protect the LED from excessive current. The correct current path is from the battery positive terminal to the resistor, from the resistor to the LED anode, from the LED cathode to the battery negative terminal, and finally to the ground reference if required.

The scheme also prepares the project for simulation by including the correct measurement positions. The ammeter is placed in series to measure current, while the voltmeter is placed in parallel with the LED to measure forward voltage. With this arrangement, the simulation can confirm whether the calculated resistor value produces a safe LED current.

A correctly completed Chapter 3 must allow another student, teacher, or examiner to rebuild the Proteus circuit exactly, without guessing any component, wire, polarity, or measurement position.

4

Simulation

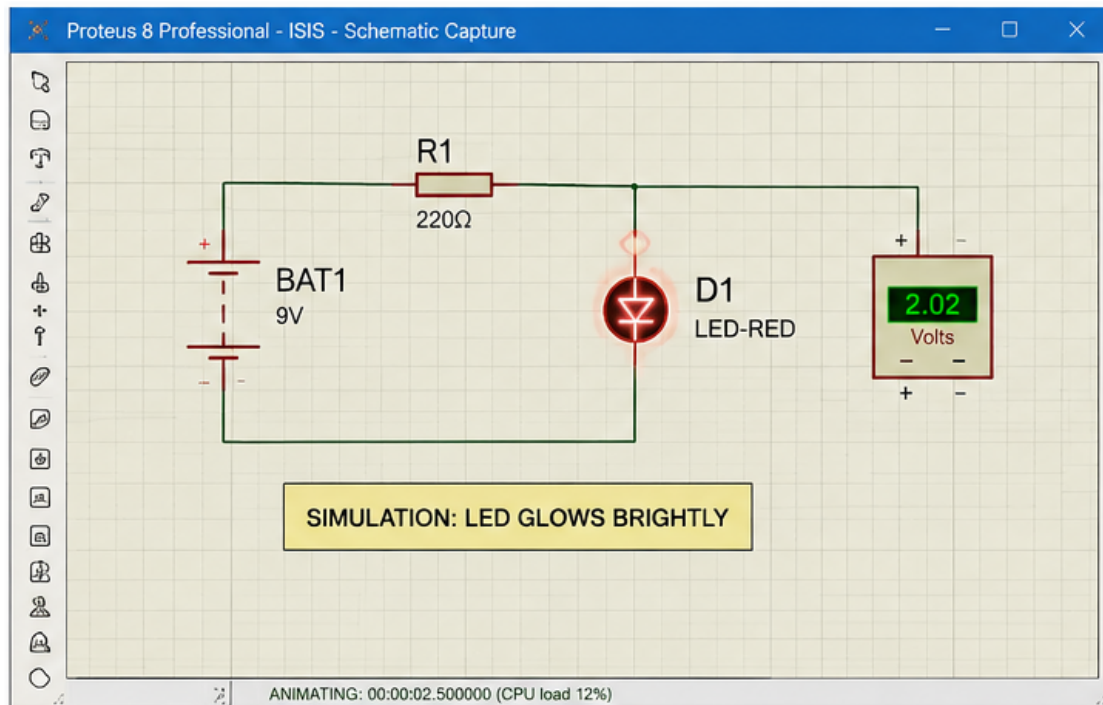


Figure 4.1: Proteus simulation result for the LED circuit

4.1 Introduction to Simulation

Simulation is an essential stage in the verification of the current-limited LED circuit because it allows the theoretical design to be tested before any physical construction is attempted. In this project, the simulation is performed in Proteus ISIS Schematic Capture in order to observe the electrical behavior of a red LED connected to a 9V battery through a series resistor. The simulation confirms whether the calculated resistor value limits the current to a safe operating range and whether the LED operates under the expected voltage and current conditions.

The purpose of this chapter is not only to show that the LED turns on, but also

to prove that the circuit behaves according to the mathematical design developed in the calculation chapter. A glowing LED alone is not sufficient evidence of correctness, because an LED may glow while still receiving excessive current. Therefore, the simulation must include measurable electrical quantities, especially circuit current, LED voltage, and resistor voltage. These values provide objective evidence that the circuit is electrically valid, safe, and consistent with Ohm's law.

In this chapter, the Proteus simulation is configured, executed, measured, and interpreted. The expected results are first calculated theoretically, and then the simulated values are compared with those expectations. This method ensures that the simulation functions as a verification tool rather than a visual demonstration only.

4.2 Purpose of the Proteus Simulation

The main purpose of the Proteus simulation is to verify that the selected resistor protects the LED by limiting the current to an acceptable value. Since the LED is a current-sensitive semiconductor component, it cannot be connected directly to a 9V battery without a current-limiting element. The resistor must absorb the remaining voltage after the LED forward voltage has been reached, and it must restrict current flow according to Ohm's law.

The simulation has five specific purposes. First, it verifies that the LED is connected with correct polarity. Second, it confirms that the resistor is placed in series with the LED and therefore controls the same current that passes through the LED. Third, it checks whether the current value is close to the target design current. Fourth, it confirms that the LED forward voltage appears across the LED terminals. Fifth, it confirms that the remaining voltage appears across the resistor.

The simulation also provides evidence that the circuit is safe under the selected design conditions. If the measured current is close to the expected value, then the calculation and schematic design are consistent. If the measured current is significantly different, the circuit must be reviewed for incorrect resistor value, wrong LED model, reversed polarity, incorrect meter placement, or missing reference connection.

4.3 Simulation Setup

The simulation setup defines the exact electrical conditions under which the circuit is tested. The Proteus circuit contains a 9V battery, a red LED, a current-limiting resistor, an ammeter, and a voltmeter. The resistor is connected in series with the LED, while the voltmeter is connected in parallel with the LED. The ammeter is placed in series so that it measures the total current flowing through both the resistor and the LED.

The simulation must be prepared carefully because incorrect component settings or incorrect instrument placement can produce misleading results. The battery voltage must match the design value, the resistor must match the selected calculated value, and the LED model must represent a normal red LED. The measuring instruments must be placed according to correct electrical measurement principles. An ammeter must always be placed in series with the load, while a voltmeter must always be placed in

parallel with the component whose voltage is being measured.

For the final recommended simulation, the resistor value is selected as 360Ω . This value is suitable because it produces a current close to the 20mA target when the red LED forward voltage is approximately 2V.

4.3.1 Battery Voltage Setting

The battery is set to 9V DC. This value represents the supply voltage of the circuit and is the source of electrical energy for the LED and resistor. The positive terminal of the battery is connected to the input side of the resistor, and the negative terminal is connected to the cathode side of the LED through the return path of the circuit.

The voltage source must remain constant during the simulation so that the measured values can be compared directly with the theoretical calculation. If the battery voltage is changed, the circuit current also changes because the resistor voltage depends on the difference between the supply voltage and the LED forward voltage.

The supply voltage is represented as:

$$V_S = 9V$$

where (V_S) is the source voltage.

4.3.2 Resistor Value Setting

The resistor is set to 360Ω for the recommended simulation. This value is selected because the calculated ideal resistance for a 9V supply, a 2V red LED forward voltage, and a 20mA target current is 350Ω . Since 350Ω may not always be used as a standard practical value, 360Ω is selected as a close and safe standard value.

The theoretical resistor value is calculated as:

$$R = \frac{V_S - V_F}{I_F}$$

Substituting the design values gives:

$$R = \frac{9V - 2V}{0.02A}$$

$$R = \frac{7V}{0.02A}$$

$$R = 350\Omega$$

The selected simulation value is therefore:

$$R_1 = 360\Omega$$

This value produces a current slightly below 20mA, which is appropriate for safe LED operation.

4.3.3 LED Model Setting

The LED model used in Proteus is a red LED. The red LED is selected because red LEDs commonly have a forward voltage near 2V under typical operating current. The LED must be connected with its anode toward the positive supply side and its cathode toward the negative supply side. This orientation allows the LED to become forward-biased and conduct current.

The LED forward voltage is represented as:

$$V_F \approx 2V$$

This value is an approximate operating value rather than an absolutely fixed voltage. In practice and in simulation, the exact LED voltage may vary slightly depending on the LED model, current level, and internal simulation parameters. However, for this educational design, 2V is an appropriate working assumption for a red LED.

4.3.4 Ammeter Placement

The ammeter is placed in series with the LED and resistor. This placement is necessary because current is measured by allowing the current to pass through the ammeter. In a series circuit, the current is the same at every point in the loop. Therefore, the ammeter may be placed between the battery and resistor, between the resistor and LED, or between the LED and battery return path, provided that it remains in series with the circuit.

The preferred placement is between the resistor and the LED because this location clearly shows that the measured current is the current entering the LED. The ammeter reading is expected to be close to 19.4mA when a 360Ω resistor is used.

The expected current is:

$$I = \frac{V_S - V_F}{R}$$

$$I = \frac{9V - 2V}{360\Omega}$$

$$I = \frac{7V}{360\Omega}$$

$$I = 0.01944A$$

$$I \approx 19.44mA$$

4.3.5 Voltmeter Placement

The voltmeter is placed in parallel with the LED in order to measure the voltage across the LED terminals. The positive terminal of the voltmeter is connected to the LED

anode, and the negative terminal of the voltmeter is connected to the LED cathode. This connection allows the voltmeter to measure the LED forward voltage during operation.

A second voltmeter may also be connected across the resistor to verify the resistor voltage. If the LED forward voltage is approximately 2V and the battery voltage is 9V, then the resistor voltage should be approximately 7V.

The expected resistor voltage is:

$$V_R = V_S - V_F$$

$$V_R = 9V - 2V$$

$$V_R = 7V$$

The expected LED voltage is:

$$V_{LED} \approx 2V$$

The expected resistor voltage is:

$$V_R \approx 7V$$

Together, these values should satisfy Kirchhoff's voltage law:

$$V_S = V_R + V_{LED}$$

$$9V \approx 7V + 2V$$

4.4 Running the Simulation

After all components have been placed and connected correctly, the Proteus simulation is started using the run command. During simulation, the LED should illuminate, the ammeter should display the circuit current, and the voltmeter should display the LED forward voltage. The simulation should be allowed to stabilize so that the displayed measurements represent steady-state operation.

The first observation is whether the LED turns on. If the LED does not turn on, the circuit must be checked for reversed LED polarity, missing wire connections, incorrect battery polarity, or an open circuit. The second observation is the ammeter reading. If the current is much higher than expected, the resistor value may be too low or the LED model may have a lower forward voltage than assumed. If the current is much lower than expected, the resistor may be too large, the LED may be reversed, or the circuit may not be fully closed.

The simulation is considered correctly executed only when the LED is forward-biased, the current is within the expected range, and the voltage readings are consistent

with the theoretical calculation.

4.5 Expected Simulation Results

The expected simulation results are derived from the calculated circuit values. Since the battery voltage is 9V, the LED forward voltage is approximately 2V, and the resistor value is 360Ω, the expected current is approximately 19.44mA. The LED voltage should be close to 2V, and the resistor voltage should be close to 7V.

These results are not independent values. They are connected by the voltage and current relationships of the circuit. The sum of the LED voltage and resistor voltage should be approximately equal to the battery voltage. The current through the resistor and the LED should be the same because the circuit is connected in series.

4.5.1 Expected LED Current

The LED current is expected to be approximately 19.44mA when a 360Ω resistor is used. This result follows from Ohm's law applied to the resistor voltage.

$$I_{LED} = \frac{V_S - V_F}{R}$$

$$I_{LED} = \frac{9V - 2V}{360\Omega}$$

$$I_{LED} = 0.01944A$$

$$I_{LED} \approx 19.44mA$$

This current is close to the 20mA target and is suitable for normal LED operation. It is also safer than using a 220Ω resistor, which would produce a higher current.

4.5.2 Expected LED Voltage

The expected LED voltage is approximately 2V because the selected component is a red LED. This voltage appears across the LED when it is forward-biased and conducting. In the simulation, the displayed LED voltage may not be exactly 2.00V because the LED model may use a nonlinear semiconductor equation rather than a fixed voltage drop. However, the measured value should remain close to the assumed forward voltage.

The expected LED voltage is:

$$V_{LED} \approx 2V$$

A value near 2V supports the conclusion that the LED is operating in its forward-biased region.

4.5.3 Expected Resistor Voltage

The resistor voltage is expected to be the remaining voltage after the LED forward voltage has been subtracted from the supply voltage.

$$V_R = V_S - V_{LED}$$

$$V_R = 9V - 2V$$

$$V_R = 7V$$

Therefore, the voltage across the resistor should be approximately 7V. This value is important because it is the voltage responsible for setting the current through the resistor according to Ohm's law.

The resistor voltage and LED voltage should satisfy the following relationship:

$$V_S = V_R + V_{LED}$$

$$9V \approx 7V + 2V$$

If this relationship is not approximately satisfied in the simulation, the circuit connections or measurement points must be checked.

4.5.4 Expected LED Brightness

The LED is expected to glow with normal visible brightness because the current is close to 20mA. The brightness should not be extremely weak, because the current is much greater than the low current produced by a 10kΩ resistor. The brightness should also not be treated as excessive, because the current is lower than the current produced by a 220Ω resistor.

The expected brightness is therefore a normal safe brightness. However, brightness is a qualitative observation and cannot replace numerical measurement. The academic judgment of the simulation must be based primarily on current and voltage readings, not only on visual appearance.

4.6 Simulation Result with Recommended Resistor

With the resistor set to 360Ω, the simulation should show a circuit current close to 19.44mA. The LED should turn on, and the voltmeter across the LED should show a value close to 2V. The resistor should carry the same current as the LED because both components are connected in series.

The expected result can be summarized as follows:

$$R_1 = 360\Omega$$

$$V_S = 9V$$

$$V_{LED} \approx 2V$$

$$V_R \approx 7V$$

$$I_{LED} \approx 19.44mA$$

This result confirms that the selected resistor is appropriate for the design objective. The LED receives enough current to operate visibly, while the resistor prevents excessive current from flowing through the LED.

If the Proteus measurement shows a small difference from the calculated current, the difference is acceptable if it is caused by the internal LED model. Real and simulated LEDs are nonlinear devices, so the forward voltage is not always exactly 2V. A small variation does not invalidate the circuit, provided that the current remains within a safe operating range.

4.7 Comparison Between Calculation and Simulation

The comparison between calculation and simulation is necessary because the calculation uses simplified assumptions, while the simulation uses component models. In the calculation, the red LED forward voltage is assumed to be approximately 2V. In Proteus, the LED model may produce a forward voltage slightly above or below this value depending on the model behavior and operating current.

The calculated current for the 360Ω resistor is:

$$I_{calculated} = 19.44mA$$

The simulated current should be close to this value. The percentage difference can be calculated as:

$$\text{Percentage Difference} = \frac{|I_{simulated} - I_{calculated}|}{I_{calculated}} \times 100$$

A small percentage difference is acceptable because the LED is not an ideal fixed-voltage component. The simulation is considered successful if the measured current remains close to the expected value and the LED operates safely.

For example, if Proteus shows 19.1mA, then:

$$\text{Percentage Difference} = \frac{|19.1mA - 19.44mA|}{19.44mA} \times 100$$

$$\text{Percentage Difference} = \frac{0.34mA}{19.44mA} \times 100$$

Percentage Difference ≈ 1.75

This difference would be acceptable for an educational circuit simulation.

4.8 Simulation Measurement Table

The simulation measurements should be recorded in a table so that the theoretical and simulated values can be compared clearly. The table must include the resistor value, calculated current, simulated current, LED voltage, resistor voltage, and LED condition.

Table 4.1: Comparison Between Calculated and Simulated Results

Parameter	Expected Value	Simulated Value	Evaluation
Supply voltage	9V	To be measured in Proteus	Must remain close to 9V
Resistor value	360 Ω	360 Ω	Correct if set properly
LED forward voltage	Approximately 2V	To be measured in Proteus	Acceptable if close to red LED forward voltage
Resistor voltage	Approximately 7V	To be measured in Proteus	Should equal supply voltage minus LED voltage
LED current	Approximately 19.44mA	To be measured in Proteus	Must be close to target current
LED brightness	Normal visible brightness	To be observed in Proteus	Must not be off or excessively bright

A second comparison table may be included to show why the 360 Ω resistor is stronger than incorrect or less suitable resistor values.

Table 4.2: Simulation Measurement Table

Resistor Value	Expected Current	Expected LED Condition	Technical Judgment
220 Ω	Approximately 31.82mA	Very bright	Not preferred for a strict 20mA design
330 Ω	Approximately 21.21mA	Bright	Acceptable only if slightly higher current is allowed
360 Ω	Approximately 19.44mA	Normal brightness	Recommended
390 Ω	Approximately 17.95mA	Safe visible brightness	Conservative alternative
10k Ω	Approximately 0.70mA	Very dim or invisible	Useful as a troubleshooting example

4.9 Interpretation of Simulation Results

The simulation results should be interpreted by checking whether the measured values agree with the theoretical behavior of the circuit. If the current is close to 19.44mA, the resistor is correctly limiting current. If the LED voltage is close to 2V, the LED is operating in its expected forward-biased region. If the resistor voltage is close to 7V, the voltage distribution is consistent with the design calculation.

The most important result is the current value. The LED is protected when the current remains within the intended operating range. The resistor performs its protective function by absorbing the voltage that is not used by the LED and by converting part of the electrical energy into heat. The power dissipated by the resistor can be estimated as:

$$P_R = V_R \times I$$

Using the expected values:

$$P_R = 7V \times 0.01944A$$

$$P_R = 0.136W$$

This result shows that a resistor rated at 0.25W is generally sufficient for the simulation condition, while a 0.5W resistor would provide a stronger safety margin in physical construction.

The LED power can also be estimated:

$$P_{LED} = V_{LED} \times I$$

$$P_{LED} = 2V \times 0.01944A$$

$$P_{LED} = 0.0389W$$

These values show that the circuit distributes voltage and power in a controlled manner. The LED receives enough power to illuminate, while the resistor prevents the LED from receiving excessive current.

4.10 Verification of LED Protection

The LED protection is verified when the simulation proves that the resistor limits the current to a safe level. In this circuit, the resistor protects the LED by controlling the amount of current that can pass through the series path. Since the LED and resistor are connected in series, the current through the resistor is the same current that passes through the LED.

The protection mechanism can be expressed as:

$$I_{LED} = I_R$$

$$I_{LED} = \frac{V_S - V_F}{R}$$

For the selected 360Ω resistor:

$$I_{LED} \approx 19.44mA$$

This value is close to the target current of 20mA. Therefore, the resistor successfully prevents the LED from receiving excessive current from the 9V battery.

The protection can also be verified by comparing the recommended resistor with a lower resistor value. If 220Ω is used, the current becomes:

$$I = \frac{9V - 2V}{220\Omega}$$

$$I = \frac{7V}{220\Omega}$$

$$I \approx 31.82mA$$

This value is higher than the intended design current and is therefore not preferred for a strict safe design. By contrast, the 360Ω resistor gives a current close to the intended target. This comparison proves that resistor selection is not decorative or optional. It is the central safety decision in the circuit.

4.11 Simulation Accuracy Discussion

The simulation is accurate for educational verification because it confirms the expected relationship between voltage, resistance, and current. However, the simulation should be understood as a model of circuit behavior rather than a perfect copy of every physical condition. Proteus uses component models, and the LED model may not behave exactly like every real red LED available in hardware.

Several factors may cause small differences between calculated and simulated results. First, the LED forward voltage may not be exactly 2V in the simulation. Second, the LED current depends on the internal semiconductor model used by Proteus. Third, the displayed meter readings may be rounded. Fourth, real physical components have tolerances, although ideal simulation components may not show those tolerances unless they are manually configured.

These limitations do not weaken the simulation if the results remain close to the calculated values. The purpose of the simulation is to verify the design principle and confirm safe circuit operation. The simulation is successful when the LED is forward-biased, the current is close to the calculated value, the voltage distribution matches the

expected relationship, and the resistor limits current to a safe operating range.

For a more complete report, the simulation should include screenshots showing the circuit before running, the circuit during operation, the ammeter reading, the LED voltage reading, and the resistor voltage reading. These images provide visual and numerical evidence that the circuit has been tested correctly.

4.12 Chapter Summary

This chapter presented the simulation of a current-limited red LED circuit powered by a 9V battery. The Proteus simulation was used to verify the theoretical calculation and confirm that the LED operates safely when a 360Ω series resistor is used. The expected current was calculated as approximately 19.44mA, which is close to the target current of 20mA. The expected LED voltage was approximately 2V, and the expected resistor voltage was approximately 7V.

The simulation confirmed the central design principle of the project: an LED must not be connected directly to a 9V battery without a current-limiting resistor. The resistor is required because it controls the current flowing through the LED and prevents excessive current from damaging the component. The chapter also showed that simulation evidence must include numerical measurements, not only visual LED brightness.

The results of this chapter prepare the report for the troubleshooting chapter, where incorrect resistor values, reversed LED polarity, open circuits, and other possible faults are examined.

5

Troubleshooting

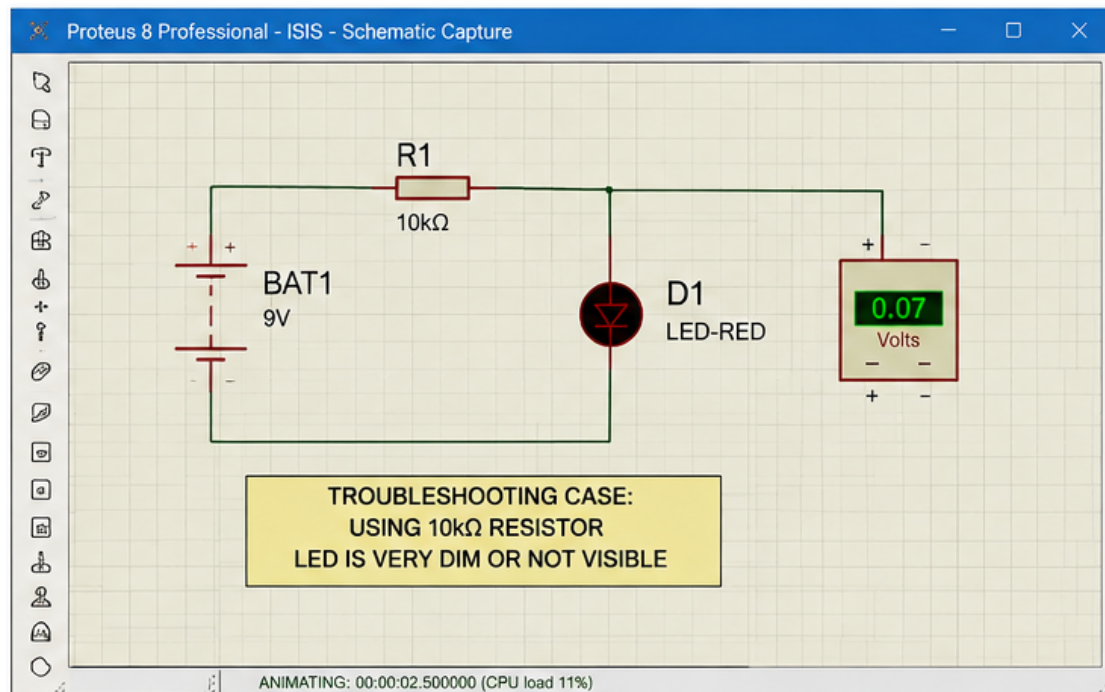


Figure 5.1: *Proteus troubleshooting case using a 10kΩ resistor*

5.1 Introduction to Troubleshooting

Troubleshooting is an essential stage in circuit design because a circuit that is theoretically correct may still fail if the component values, wiring connections, polarity, measurement placement, or simulation settings are incorrect. In the current limited LED circuit, the main objective is to ensure that the LED receives enough current to illuminate while remaining protected from excessive current. Since an LED is a polarity sensitive semiconductor device, its behavior depends not only on the supply voltage but also on the direction of connection, the value of the series resistor, and the continuity of the circuit path.

This chapter examines the most important fault conditions that may occur in the 9V battery, resistor, and LED circuit. Each fault is analyzed through its cause, electrical effect, expected simulation observation, and correction method. The purpose is not only to identify visible problems, such as an LED that does not glow, but also to explain the electrical reason behind each problem. A correct troubleshooting process must move from observation to measurement, from measurement to cause, and from cause to correction.

The circuit under investigation is based on a 9V DC supply, a red LED with an assumed forward voltage of approximately 2V, and a current limiting resistor connected in series. The expected design current is approximately 20mA. Therefore, the normal circuit must satisfy the following design relationship:

$$R = \frac{V_S - V_F}{I_F}$$

where (V_S) is the supply voltage, (V_F) is the LED forward voltage, and (I_F) is the desired LED forward current. When the circuit does not behave as expected, troubleshooting must check whether this relationship has been violated by an incorrect resistor value, wrong polarity, missing connection, or incorrect measurement method.

5.2 Purpose of Troubleshooting in Circuit Design

The purpose of troubleshooting in this project is to verify whether the LED circuit performs according to the theoretical calculation and the intended Proteus schematic. Troubleshooting is necessary because the apparent condition of the LED is not sufficient evidence of correct circuit behavior. An LED may glow brightly while receiving excessive current, or it may remain dark because the resistor is too large, the LED is reversed, the circuit is open, or the measuring instruments are connected incorrectly.

In this project, troubleshooting has five main purposes. First, it confirms that the current through the LED remains within a safe range. Second, it identifies whether the resistor value produces the expected current. Third, it checks whether the LED polarity is correct. Fourth, it verifies that the circuit forms a complete closed loop from the battery positive terminal to the battery negative terminal. Fifth, it ensures that the ammeter and voltmeter are connected in the correct positions so that the simulation measurements are meaningful.

A proper troubleshooting method must follow a fixed technical sequence. The first step is visual inspection of the circuit layout. The second step is checking component values. The third step is verifying polarity. The fourth step is measuring voltage and current. The fifth step is comparing measured values with calculated values. The final step is applying the correction and confirming that the corrected circuit behaves as expected.

For the LED circuit, the central troubleshooting equation is:

$$I = \frac{V_S - V_F}{R}$$

This equation allows the designer to determine whether the selected resistor value

produces too much current, too little current, or an acceptable current. Therefore, troubleshooting in this chapter is not based on guessing. It is based on circuit law, measured simulation behavior, and logical fault isolation.

5.3 Common Faults in the LED Circuit

Several common faults can occur in the LED circuit. The first fault is using a resistor value that is too large. In this case, the circuit is safe, but the current becomes too low and the LED may appear very dim or completely off. The second fault is using a resistor value that is too small. In this case, the LED may glow brightly, but the current may exceed the desired safe operating current. The third fault is connecting the LED directly to the battery without any resistor. This is a serious design error because the LED has no reliable current limiting protection.

The fourth fault is reversing the LED polarity. Since an LED conducts mainly in the forward direction, reversed polarity prevents normal conduction. In a simulation, the LED usually remains off. In a physical circuit, excessive reverse voltage may also damage the LED. The fifth fault is an open circuit. If one wire is missing or disconnected, there is no complete path for current, so the current becomes zero and the LED remains off.

The sixth fault is wrong battery connection. If the supply polarity is reversed, the LED becomes reverse biased. If the supply value is set incorrectly, the current may become too high or too low. The seventh fault is wrong meter connection. An ammeter must be placed in series because it measures current through the circuit path. A voltmeter must be placed in parallel with the component whose voltage is being measured. If these instruments are connected incorrectly, the result may be misleading or may change the circuit behavior.

Therefore, the common faults in this circuit can be grouped into four categories: value faults, polarity faults, continuity faults, and measurement faults. Value faults involve incorrect resistor or supply values. Polarity faults involve reversed LED or battery orientation. Continuity faults involve broken wires or incomplete loops. Measurement faults involve incorrect use of the ammeter or voltmeter.

5.4 Troubleshooting Case 1: Using a 10k Ω Resistor

Using a 10k Ω resistor instead of the calculated current limiting resistor is a common example of excessive resistance in an LED circuit. This fault does not usually destroy the LED, because the current becomes very small. However, it prevents the LED from operating at the intended brightness. The LED may appear extremely dim or completely invisible to the observer.

For this project, the assumed supply voltage is 9V and the assumed red LED forward voltage is approximately 2V. Therefore, the resistor must drop the remaining voltage:

$$V_R = V_S - V_F$$

$$V_R = 9V - 2V = 7V$$

If the resistor is 10kΩ, the circuit current becomes:

$$I = \frac{7V}{10000\Omega}$$

$$I = 0.0007A$$

$$I = 0.7mA$$

A current of 0.7mA is far below the normal target current of approximately 20mA. Because LED brightness is strongly related to forward current, the LED receives insufficient current and therefore produces very weak light.

5.4.1 Cause

The cause of this fault is the selection of a resistor value that is too large for the required LED current. The resistor limits current by opposing current flow. When the resistance is increased from the recommended range to 10kΩ, the circuit current decreases drastically.

The intended current is approximately:

$$I_F = 20mA$$

The faulty current is:

$$I = 0.7mA$$

The percentage of the intended current is:

$$\frac{0.7mA}{20mA} \times 100 = 3.5$$

This means the LED receives only about 3.5 percent of the intended design current. The circuit is not damaged, but it fails to meet the functional requirement of visible brightness.

5.4.2 Calculation

The calculation confirms the fault clearly. For a 9V battery and a red LED forward voltage of approximately 2V:

$$V_R = 7V$$

With ($R = 10000\Omega$):

$$I = \frac{V_R}{R}$$

$$I = \frac{7V}{10000\Omega}$$

$$I = 0.0007A$$

$$I = 0.7mA$$

The power dissipated by the resistor is:

$$P_R = I^2 R$$

$$P_R = (0.0007A)^2 \times 10000\Omega$$

$$P_R = 0.0049W$$

$$P_R = 4.9mW$$

The resistor power is very small, so overheating is not expected. The main issue is not safety but insufficient current.

5.4.3 Simulation Observation

In Proteus, the LED is expected to appear very dim or may not visibly glow. The ammeter should show a current close to 0.7mA, depending on the exact LED model used in the simulation. The voltmeter across the LED may still show a forward voltage, but the forward current is too small to produce normal brightness.

The expected simulation condition is therefore:

The circuit is electrically complete.

The LED is connected in the correct direction.

The resistor value is too high.

The current is too low.

The LED brightness is weak or invisible.

This case proves that a circuit can be correctly wired but still function poorly because the component value is unsuitable.

5.4.4 Correction

The correction is to replace the 10kΩ resistor with a resistor value that produces a current close to the intended design current. For a 9V supply, a red LED forward voltage of approximately 2V, and a target current of 20mA, the calculated resistance is:

$$R = \frac{9V - 2V}{0.02A}$$

$$R = 350\Omega$$

A practical standard value such as 360Ω or 390Ω should be used for a safer design. After replacing the 10kΩ resistor, the simulation should be repeated. The ammeter should then show a current close to the expected range, and the LED should glow normally.

5.5 Troubleshooting Case 2: Using a 220Ω Resistor

Using a 220Ω resistor creates the opposite problem. In this case, the LED receives more current than the strict 20mA design target. Although the LED may glow brightly, this does not mean the circuit is correctly designed. A bright LED can hide an overcurrent condition. Therefore, the 220Ω case must be treated as a technical warning rather than a successful design result.

For a 9V battery and a red LED forward voltage of approximately 2V, the resistor voltage remains:

$$V_R = 7V$$

With a 220Ω resistor, the current is:

$$I = \frac{7V}{220\Omega}$$

$$I = 0.0318A$$

$$I = 31.8mA$$

This current is significantly higher than 20mA. Therefore, 220Ω is not the most defensible resistor value for a strict 20mA design.

5.5.1 Cause

The cause of this fault is the selection of a resistor value that is too small. A smaller resistor allows more current to pass through the LED. Since the LED is a current sensitive device, excessive current can increase heat and reduce component reliability.

The intended current is:

$$20mA$$

The current produced by 220Ω is:

$$31.8mA$$

The excess current above the 20mA target is:

$$31.8mA - 20mA = 11.8mA$$

The percentage increase above the target is:

$$\frac{11.8mA}{20mA} \times 100 = 59$$

This means the LED current is approximately 59 percent higher than the intended design current.

5.5.2 Calculation

The calculation is as follows:

$$V_R = V_S - V_F$$

$$V_R = 9V - 2V = 7V$$

$$I = \frac{V_R}{R}$$

$$I = \frac{7V}{220\Omega}$$

$$I = 0.0318A$$

$$I = 31.8mA$$

The resistor power is:

$$P_R = I^2 R$$

$$P_R = (0.0318A)^2 \times 220\Omega$$

$$P_R \approx 0.222W$$

This power is close to the common 0.25W resistor rating. In a physical circuit, this leaves little safety margin. A stronger design should avoid operating so close to the resistor rating and should reduce the LED current to the intended value.

5.5.3 Simulation Observation

In Proteus, the LED will probably glow brightly because the current is higher than the normal target. The ammeter should show a value near 31.8mA, depending on the LED model. The LED may appear visually successful, but the measured current shows that the design is not ideal for a 20mA target.

The expected simulation condition is therefore:

The LED glows.

The circuit is complete.

The resistor value is too low.

The current is higher than the intended design current.

The result is not a properly controlled 20mA LED circuit.

This case proves that visible brightness must not be accepted as the only success criterion. Measurement is required.

5.5.4 Correction

The correction is to replace the 220Ω resistor with a more suitable resistor value. The calculated resistance for a 20mA design is:

$$350\Omega$$

Since 350Ω may not always be used as a common selected value, 360Ω or 390Ω is preferred. With 360Ω:

$$I = \frac{7V}{360\Omega}$$

$$I = 19.4mA$$

With 390Ω:

$$I = \frac{7V}{390\Omega}$$

$$I = 17.9mA$$

Both values provide safer current control than 220Ω. The simulation should be repeated after replacement to confirm that the current is now within the intended range.

5.6 Troubleshooting Case 3: LED Connected Without Resistor

Connecting the LED directly to the 9V battery is the most serious design error in this project. An LED must not be connected directly to a voltage source without current limiting. The LED forward voltage is much lower than the battery voltage, and once the LED conducts, current can rise rapidly. Without a resistor, the current is limited only by the internal resistance of the battery, the LED characteristics, and the wiring

resistance. These are not reliable design controls.

A direct connection may cause excessive LED current, overheating, permanent LED damage, battery stress, or inaccurate simulation behavior. In a real circuit, the LED may flash briefly and then fail. In simulation, the result may depend on the exact component model, but the design principle remains invalid.

5.6.1 Cause

The cause of this fault is the absence of a current limiting resistor. The LED is a semiconductor device with a nonlinear voltage current relationship. It does not behave like a resistor that naturally limits current in a predictable linear manner. Therefore, a direct connection to 9V gives the circuit no deliberate current control.

The safe design requires:

$$R = \frac{V_S - V_F}{I_F}$$

If ($R = 0\Omega$), the equation no longer gives a safe finite design current. The remaining voltage after the LED forward voltage has no intentional resistance across which it can be safely dropped.

5.6.2 Risk

The main risk is excessive current. In the intended design, the resistor absorbs approximately 7V and limits the current to approximately 20mA. Without the resistor, the circuit attempts to force a much larger current through the LED. This can exceed the LED current rating and damage the component.

The thermal risk is also important. Electrical power in the LED is approximately:

$$P_{LED} = V_F \times I_F$$

If the current rises beyond the safe value, LED power also rises. Higher power creates heat. Higher heat can further change the LED behavior and reduce reliability. Therefore, the missing resistor is not a minor error. It removes the main protective component from the circuit.

5.6.3 Correction

The correction is to disconnect the direct LED battery connection and insert a resistor in series with the LED. The corrected path must be:

Battery positive terminal to resistor.
Resistor to LED anode.
LED cathode to battery negative terminal.

The corrected resistor value should be calculated before simulation. For the assumed values:

$$R = \frac{9V - 2V}{0.02A}$$

$$R = 350\Omega$$

A practical corrected value is 360Ω or 390Ω . The circuit should not be tested physically without the resistor.

5.7 Troubleshooting Case 4: Reversed LED Polarity

An LED conducts when it is forward biased. This means the anode must be connected toward the positive side of the circuit, and the cathode must be connected toward the negative side. If the LED is reversed, it becomes reverse biased and does not conduct normally. As a result, the LED remains off even if the resistor value and battery voltage are correct.

This is one of the most common beginner errors because the circuit may appear visually complete while being electrically blocked by the reversed semiconductor direction. Therefore, polarity checking is a necessary part of troubleshooting.

5.7.1 Cause

The cause of this fault is incorrect LED orientation. The LED anode and cathode have been exchanged. In the intended circuit, current must flow from the positive terminal of the battery through the resistor, then into the LED anode, through the LED, and out of the cathode toward the battery negative terminal.

If the LED is reversed, the current path is blocked because the LED does not allow normal forward conduction. The circuit may show nearly zero current, except for very small leakage current in a real component or model dependent leakage in simulation.

5.7.2 Simulation Observation

In Proteus, the LED will remain off when reversed. The ammeter should show zero or a very small current. The voltmeter may show voltage across the LED, but that voltage is reverse voltage rather than normal forward voltage. This is an important distinction. A voltage reading alone does not prove correct LED operation. The direction of polarity and the measured current must also be checked.

The expected simulation condition is therefore:

- The battery is present.
- The resistor may be correct.
- The LED orientation is incorrect.
- The LED does not glow.
- The measured current is zero or extremely small.

5.7.3 Correction

The correction is to reverse the LED orientation. The anode must be connected to the resistor side coming from the battery positive terminal. The cathode must be connected toward the battery negative terminal. In Proteus, the LED symbol direction should be checked carefully before running the simulation again.

After correction, the simulation should show normal forward current. If the resistor is 360Ω , the current should be approximately:

$$I = \frac{7V}{360\Omega}$$

$$I = 19.4mA$$

If the resistor is 390Ω , the current should be approximately:

$$I = \frac{7V}{390\Omega}$$

$$I = 17.9mA$$

The LED should then glow normally.

5.8 Troubleshooting Case 5: Open Circuit

An open circuit occurs when the current path is broken. This can happen if a wire is missing, a component terminal is not connected, a node is not joined correctly, or the battery negative terminal is not connected to the return path. In Proteus, a visually close wire may still fail to connect if the endpoint is not properly attached to the component pin.

In an open circuit, no continuous path exists from the battery positive terminal to the battery negative terminal. Therefore, current cannot flow through the LED. The LED remains off even if all component values are correct.

5.8.1 Cause

The cause of an open circuit is a missing or broken electrical connection. The circuit requires a closed loop. The required loop is:

Battery positive terminal to resistor.

Resistor to LED anode.

LED cathode to battery negative terminal.

If any part of this loop is disconnected, the circuit current becomes:

$$I = 0A$$

This is true regardless of the resistor value because current cannot flow through an incomplete path.

5.8.2 Simulation Observation

In Proteus, the LED will remain off. The ammeter will show zero current. A voltmeter placed across the battery will still show 9V, but the presence of battery voltage does not prove that the circuit path is complete. A voltmeter placed across an open point may show unexpected voltage depending on the location of the break.

The expected simulation condition is therefore:

The supply voltage exists.

The circuit path is incomplete.

The current is zero.

The LED remains off.

The fault is caused by missing continuity.

5.8.3 Correction

The correction is to inspect every connection point and restore the closed loop. The designer should verify that each wire endpoint is physically attached to the intended component terminal in Proteus. The corrected loop must connect the battery positive terminal, resistor, LED anode, LED cathode, and battery negative terminal without interruption.

After the wire is corrected, the simulation should be run again. If the resistor value and LED polarity are correct, the LED should glow and the ammeter should show the expected current.

5.9 Troubleshooting Case 6: Wrong Battery Connection

Wrong battery connection means that the supply is connected with incorrect polarity, incorrect voltage value, or incorrect reference connection. In the LED circuit, battery polarity is especially important because the LED conducts only in the forward direction. If the battery is reversed, the LED becomes reverse biased and does not glow. If the battery value is entered incorrectly, the current calculation becomes invalid.

For example, if the battery is set to a value lower than the LED forward voltage, the LED may not turn on. If the battery is set much higher than 9V without changing the resistor, the current may become excessive. Therefore, troubleshooting must confirm both the polarity and the voltage value of the battery.

5.9.1 Cause

The cause of this fault is incorrect supply configuration. The battery may be connected in reverse, set to the wrong voltage, or not connected properly to the circuit reference. Since the current calculation depends directly on the supply voltage, any wrong battery value changes the entire circuit behavior.

The correct design assumes:

$$V_S = 9V$$

If the supply voltage is not 9V, then the current must be recalculated:

$$I = \frac{V_S - V_F}{R}$$

For example, if ($V_S = 12V$) and ($R = 360\Omega$):

$$I = \frac{12V - 2V}{360\Omega}$$

$$I = 27.8mA$$

This is higher than the intended 20mA target. Therefore, the supply value must not be changed without recalculating the resistor.

5.9.2 Simulation Observation

If the battery polarity is reversed, the LED remains off because it is reverse biased. The current will be zero or extremely small. If the battery voltage is too low, the LED may be dim or off. If the battery voltage is too high, the LED may be overly bright and the current may exceed the intended value.

The expected observations depend on the specific battery error:

With reversed battery polarity, the LED is off.

With too low battery voltage, the LED is dim or off.

With too high battery voltage, the LED current becomes excessive.

With missing reference, the simulation may fail or behave incorrectly.

5.9.3 Correction

The correction is to verify that the battery positive terminal is connected toward the resistor and LED anode side, while the battery negative terminal is connected toward the LED cathode side. The battery value must be set to 9V. If a different supply voltage is intentionally used, the resistor must be recalculated.

For a general supply voltage, the corrected resistor equation is:

$$R = \frac{V_S - V_F}{I_F}$$

This confirms that the resistor value is not universal. It depends on the selected supply voltage, the LED forward voltage, and the desired current.

5.10 Troubleshooting Case 7: Wrong Meter Connection

Measurement errors can make a correct circuit appear incorrect or can make an incorrect circuit appear acceptable. In this project, the ammeter and voltmeter have different functions and must be connected differently. The ammeter measures current and must be placed in series with the circuit. The voltmeter measures potential difference and must be placed in parallel with the component being measured.

Incorrect meter placement is a serious troubleshooting issue because it affects the validity of the simulation evidence. A report that includes wrong measurements cannot prove that the circuit is working correctly.

5.10.1 Cause

The cause of this fault is misunderstanding the function of each measuring instrument. An ammeter has very low internal resistance and is designed to be inserted into the current path. If it is connected in parallel across the battery or across a component, it may create a very low resistance path that distorts the circuit.

A voltmeter has very high internal resistance and is designed to measure voltage across two points. If it is placed in series, it may block or severely reduce current flow. Therefore, wrong meter placement changes the circuit and produces invalid results.

The correct connections are:

The ammeter must be connected in series with the LED and resistor. The voltmeter must be connected in parallel with the LED to measure LED voltage. A second voltmeter may be connected in parallel with the resistor to measure resistor voltage.

5.10.2 Measurement Error

If the ammeter is connected in parallel with the LED or battery, the measured result is not the actual LED current. The circuit may behave as if a short path has been introduced. If the voltmeter is connected in series, the LED may not glow because the voltmeter's high resistance prevents normal current flow.

Correct current measurement requires the same current to pass through the ammeter that passes through the LED. Therefore, the ammeter must interrupt the circuit path and become part of the series loop. Correct voltage measurement requires measuring the difference between two nodes without significantly changing the circuit current. Therefore, the voltmeter must be connected across the component.

5.10.3 Correction

The correction is to remove the wrongly placed meter and reconnect it according to its function. The ammeter should be inserted into the main current path. A suitable location is between the resistor and the LED anode, or between the battery positive terminal and the resistor. The voltmeter should be connected across the LED, with its positive terminal connected to the LED anode and its negative terminal connected to the LED cathode.

After correcting the meter positions, the simulation must be repeated. The current reading should match the calculated value, and the LED voltage should be close to the expected forward voltage for the selected LED model.

Table 5.1: *Fault Diagnosis and Correction Table*

Fault condition	Main cause	Expected current	Expected LED condition	Correct action
10k Ω resistor used	Resistance is too high	Approximately 0.7mA	Very dim or off	Replace with 360 Ω or 390 Ω
220 Ω resistor used	Resistance is too low	Approximately 31.8mA	Bright but current is too high	Replace with 360 Ω or 390 Ω
No resistor used	No current limiting element	Excessive and uncontrolled	Possible damage	Add a series resistor
LED reversed	Incorrect polarity	Zero or very small	Off	Reverse the LED orientation
Open circuit	Broken current path	0A	Off	Reconnect the broken wire
Battery reversed	Wrong supply polarity	Zero or very small	Off	Correct battery polarity
Battery voltage too high	Wrong voltage setting	Higher than intended	Too bright or unsafe	Recalculate resistor
Battery voltage too low	Wrong voltage setting	Lower than intended	Dim or off	Set battery to 9V or recalculate
Ammeter connected incorrectly	Meter not placed in series	Invalid reading	Misleading result	Insert ammeter in series
Voltmeter connected incorrectly	Meter not placed in parallel	Invalid reading	Misleading result	Connect voltmeter in parallel

5.11 Fault Diagnosis Table

This table shows that every fault must be diagnosed by connecting the visible symptom to a measurable electrical cause. The LED being off does not always mean the LED is damaged. It may indicate excessive resistance, reversed polarity, an open circuit, or incorrect supply connection. Similarly, the LED being bright does not always mean the circuit is safe. It may indicate excessive current.

5.12 Final Troubleshooting Checklist

The final troubleshooting checklist provides a systematic method for verifying the complete circuit. The first check is the supply voltage. The battery must be set to 9V. The second check is the resistor value. The resistor should be selected according to the calculation, with 360Ω or 390Ω preferred for the assumed LED parameters. The third check is LED polarity. The anode must face the positive side through the resistor, and the cathode must face the negative side.

The fourth check is circuit continuity. The circuit must form a closed loop from the battery positive terminal through the resistor and LED back to the battery negative terminal. The fifth check is meter placement. The ammeter must be in series, and the voltmeter must be in parallel with the measured component. The sixth check is the expected current. The measured current should be close to the calculated value. The seventh check is LED brightness. The LED should glow normally, but brightness must be interpreted together with current measurement.

The checklist may be written as follows:

- Confirm that the battery is set to 9V.
- Confirm that the resistor is not 10kΩ unless testing the dim fault case.
- Confirm that the resistor is not 220Ω for the final safe 20mA design.
- Confirm that 360Ω or 390Ω is used for the final recommended circuit.
- Confirm that the resistor is connected in series with the LED.
- Confirm that the LED anode is connected toward the positive side.
- Confirm that the LED cathode is connected toward the negative side.
- Confirm that the circuit path is closed.
- Confirm that no wire bypasses the LED.
- Confirm that no wire bypasses the resistor.
- Confirm that the ammeter is placed in series.
- Confirm that the voltmeter is placed in parallel.
- Confirm that the measured current matches the expected current.
- Confirm that the LED voltage is close to its expected forward voltage.
- Confirm that the resistor voltage is approximately the supply voltage minus the LED forward voltage.

The final verification calculation for a 360Ω resistor is:

$$I = \frac{9V - 2V}{360\Omega}$$

$$I = 19.4mA$$

The final verification calculation for a 390Ω resistor is:

$$I = \frac{9V - 2V}{390\Omega}$$

$$I = 17.9mA$$

Therefore, if the measured current is close to these values, the circuit is functioning correctly.

5.13 Chapter Summary

This chapter presented a complete troubleshooting analysis of the 9V LED circuit with a current limiting resistor. The analysis showed that circuit failure can result from incorrect resistor value, missing resistor protection, reversed LED polarity, open circuit wiring, wrong battery connection, or incorrect meter placement. Each fault was examined through its cause, calculation, simulation observation, and correction.

The $10k\Omega$ resistor case demonstrated that excessive resistance reduces the current to approximately 0.7mA, causing the LED to appear very dim or off. The 220Ω resistor case demonstrated that low resistance increases the current to approximately 31.8mA, which is higher than the intended 20mA design target. The no resistor case showed that removing the current limiting element creates an unsafe circuit. The reversed LED case showed that polarity is essential for LED operation. The open circuit case showed that current cannot flow without a complete loop. The wrong battery connection case showed that supply polarity and voltage must match the design assumptions. The wrong meter connection case showed that measurement instruments must be connected correctly to produce valid results.

The final conclusion of this chapter is that troubleshooting must be based on measurement and calculation, not appearance alone. A correct LED circuit must satisfy four conditions: the resistor value must be suitable, the LED polarity must be correct, the circuit path must be complete, and the measured current must match the theoretical prediction. For the assumed 9V supply and 2V red LED, the most reliable final resistor choices are 360Ω or 390Ω , because they keep the LED current close to the desired safe operating range.

6

Questions

6.1 Introduction to Review Questions

This chapter presents a structured set of review questions designed to evaluate the reader's understanding of the complete LED current limiting circuit. The questions are arranged according to the main technical parts of the report, beginning with conceptual understanding and continuing toward numerical calculation, circuit interpretation, Proteus simulation, and fault diagnosis.

The purpose of this chapter is not only to test memory, but also to confirm whether the reader can apply the principles of voltage, current, resistance, polarity, and circuit protection to a real circuit design. Since the project is based on connecting a red LED safely to a 9V battery, the reader must be able to explain why the LED requires a resistor, how the resistor value is calculated, how the circuit is wired in Proteus, how the simulation results are verified, and how incorrect circuit conditions are diagnosed.

The questions in this chapter are therefore divided into different categories. Description based questions test the general purpose of the circuit. Calculation based questions test mathematical correctness. Scheme based questions test wiring accuracy. Simulation based questions test the interpretation of measured results. Troubleshooting based questions test the ability to identify and correct faults. The final checklist confirms whether the student has achieved the expected learning outcomes of the project.

6.2 Description Based Questions

The description based questions assess whether the reader understands the purpose, components, and working principle of the LED protection circuit.

What is the main purpose of the LED current limiting circuit?

Why should an LED not be connected directly to a 9V battery?

What is the function of the resistor in this circuit?

Why is the circuit called a current limiting circuit?

What are the main components used in the project?

- What is the role of the 9V battery?
- What is the role of the red LED?
- What is the role of the connecting wires?
- Why is Proteus used in this project?
- What is meant by LED forward voltage?
- What is meant by LED forward current?
- Why is the resistor considered a protective component?
- What happens to an LED when excessive current passes through it?
- Why is circuit protection important in basic electronics?
- What practical skill does this project teach?

These questions are important because the reader must first understand the physical meaning of the circuit before attempting calculation or simulation. A student who cannot explain why the resistor is needed cannot correctly justify the design, even if the numerical result is memorized.

6.3 Calculation Based Questions

The calculation based questions assess whether the reader can correctly apply Ohm's law to determine the safe resistor value for the LED circuit.

State Ohm's law.

Write the formula used to calculate the current limiting resistor for an LED circuit.

Why is the LED forward voltage subtracted from the battery voltage?

Given a 9V battery, a red LED forward voltage of 2V, and a desired current of 20mA, calculate the required resistor value.

Convert 20mA into amperes.

Calculate the voltage across the resistor when the supply voltage is 9V and the LED forward voltage is 2V.

Calculate the current when the resistor value is 220Ω.

Calculate the current when the resistor value is 330Ω.

Calculate the current when the resistor value is 360Ω.

Calculate the current when the resistor value is 390Ω.

Calculate the current when the resistor value is 10kΩ.

Which resistor value is closest to the calculated ideal value?

Which resistor value gives a safer current, 360Ω or 390Ω?

Calculate the power dissipated by the resistor when the resistor voltage is 7V and the current is 20mA.

Why should the resistor power rating be considered?

The central calculation is:

$$R = \frac{V_S - V_F}{I_F}$$

where (V_S) is the supply voltage, (V_F) is the LED forward voltage, and (I_F) is the desired LED forward current.

For this project:

$$V_S = 9V$$

$$V_F = 2V$$

$$I_F = 20mA = 0.02A$$

Therefore:

$$R = \frac{9V - 2V}{0.02A}$$

$$R = \frac{7V}{0.02A}$$

$$R = 350\Omega$$

This calculation proves that the resistor value should be selected near 350Ω . A standard value of 360Ω is very close to the calculated value, while 390Ω gives a slightly lower and safer current.

6.4 Scheme Based Questions

The scheme based questions assess whether the reader can understand and reproduce the correct Proteus circuit wiring.

Which terminal of the battery should be connected to the resistor?

Which terminal of the resistor should be connected to the LED?

Which terminal of the LED is the anode?

Which terminal of the LED is the cathode?

Should the LED anode face the positive side or the negative side of the battery?

Should the LED cathode return toward the positive side or the negative side of the battery?

Why must the resistor be connected in series with the LED?

Can the resistor be placed after the LED instead of before the LED?

What is the correct current path in the circuit?

Where should the ammeter be placed?

Where should the voltmeter be placed to measure LED voltage?

Why is the voltmeter connected in parallel?

Why is the ammeter connected in series?

What happens if the LED is reversed?

What happens if one wire is missing?

The exact current path must be understood as follows:

Battery positive terminal connects to the first terminal of the resistor. The second terminal of the resistor connects to the LED anode. The LED cathode connects back to the battery negative terminal. This forms one complete closed loop. Since the resistor and LED are in series, the same current passes through both components.

The correct wiring order is:

Battery positive → *Resistor* → *LED anode* →
LED cathode → *Battery negative*

This wiring is correct because the LED is forward biased and the resistor is placed in the same path as the LED current.

6.5 Simulation Based Questions

The simulation based questions assess whether the reader can interpret Proteus results correctly and compare simulation measurements with theoretical calculations.

What is the purpose of simulating the circuit in Proteus?

What value should the battery have in the simulation?

What resistor value should be used for the recommended circuit?

What current should be expected when a 360Ω resistor is used?

What current should be expected when a 390Ω resistor is used?

What voltage should appear approximately across the red LED?

What voltage should appear approximately across the resistor?

Why is the LED brightness alone not enough to prove the circuit is correct?

Which instrument is used to measure current in Proteus?

Which instrument is used to measure voltage in Proteus?

Where should the current meter be placed?

Where should the voltage meter be placed?

What should the simulation show if the LED is correctly connected?

What should the simulation show if the resistor value is too high?

What should the simulation show if the LED polarity is reversed?

For a 360Ω resistor, the expected current is:

$$I = \frac{9V - 2V}{360\Omega}$$

$$I = \frac{7V}{360\Omega}$$

$$I = 0.0194A$$

$$I = 19.4mA$$

For a 390Ω resistor, the expected current is:

$$I = \frac{9V - 2V}{390\Omega}$$

$$I = \frac{7V}{390\Omega}$$

$$I = 0.0179A$$

$$I = 17.9mA$$

The simulation should therefore show a current close to these values, depending on the exact LED model used in Proteus. The LED voltage should be approximately 2V for a red LED, and the remaining voltage should appear across the resistor.

6.6 Troubleshooting Based Questions

The troubleshooting based questions assess whether the reader can identify, explain, and correct common faults in the LED circuit.

What happens if a $10k\Omega$ resistor is used instead of the recommended resistor?

Why does a $10k\Omega$ resistor make the LED very dim or invisible?

What happens if a 220Ω resistor is used?

Why can a 220Ω resistor produce excessive current for a strict 20mA design?

What happens if the LED is connected without a resistor?

What happens if the LED is reversed?

What happens if the battery is connected with incorrect polarity?

What happens if the circuit has an open wire?

What happens if the ammeter is connected incorrectly?

What happens if the voltmeter is connected incorrectly?

What should be checked first if the LED does not glow?

What should be checked after confirming the battery voltage?

How can Proteus help identify a current problem?

How can the calculated current help explain a dim LED?

Why should troubleshooting include both observation and calculation?

For the $10k\Omega$ case:

$$I = \frac{9V - 2V}{10000\Omega}$$

$$I = \frac{7V}{10000\Omega}$$

$$I = 0.0007A$$

$$I = 0.7mA$$

This current is far below the normal operating current of a typical red LED. The LED may therefore appear very dim or completely invisible.

For the 220Ω case:

$$I = \frac{9V - 2V}{220\Omega}$$

$$I = \frac{7V}{220\Omega}$$

$$I = 0.0318A$$

$$I = 31.8mA$$

This current is higher than the 20mA design target. Therefore, 220Ω should not be treated as the strongest recommended value for a strict safe design using the stated assumptions.

6.7 Multiple Choice Questions

The multiple choice questions provide a direct method for checking fundamental understanding.

What is the main purpose of the resistor in the LED circuit?

- A. To increase the battery voltage
- B. To limit current through the LED
- C. To reverse the LED polarity
- D. To remove the need for a battery

Correct answer: B

What is the correct formula for the LED current limiting resistor?

- A. $(R = V_S \times I_F)$
- B. $(R = \frac{I_F}{V_S - V_F})$
- C. $(R = \frac{V_S - V_F}{I_F})$
- D. $(R = V_F \times V_S)$

Correct answer: C

What is the approximate resistor voltage when a 9V battery powers a red LED with a 2V forward voltage?

- A. 2V
- B. 7V
- C. 9V
- D. 11V

Correct answer: B

What is the calculated resistor value for 9V supply, 2V LED forward voltage, and 20mA current?

- A. 100 Ω
- B. 220 Ω
- C. 350 Ω
- D. 10k Ω

Correct answer: C

What happens when a 10k Ω resistor is used in this circuit?

- A. The LED receives too much current
- B. The LED becomes very dim or invisible
- C. The battery voltage increases
- D. The resistor stops working as a resistor

Correct answer: B

Where should the ammeter be connected?

- A. In series with the circuit
- B. In parallel with the LED
- C. Across the battery only
- D. Across the resistor only

Correct answer: A

Where should the voltmeter be connected to measure LED voltage?

- A. In series with the LED
- B. In parallel with the LED
- C. In series with the battery
- D. In place of the resistor

Correct answer: B

What happens if the LED is reversed?

- A. It becomes brighter
- B. It works normally
- C. It does not glow
- D. It changes the resistor value

Correct answer: C

Which resistor gives a current closest to 20mA in this project?

- A. 10k Ω
- B. 220 Ω
- C. 360 Ω
- D. 1 Ω

Correct answer: C

Why is simulation useful?

- A. It replaces all calculation permanently
- B. It allows the circuit to be tested before physical construction
- C. It removes the need for correct wiring
- D. It makes polarity unimportant

Correct answer: B

6.8 Short Answer Questions

The short answer questions require clear written explanations rather than selecting from options.

Explain why an LED needs a current limiting resistor.

Explain why the resistor voltage is calculated as $(V_S - V_F)$.

Explain why a 360Ω resistor is a strong choice for this circuit.

Explain why a 390Ω resistor is safer than a 330Ω resistor.

Explain why 220Ω is not the best value for a strict 20mA target.

Explain why a $10k\Omega$ resistor makes the LED very dim.

Explain how the LED should be connected in the circuit.

Explain the difference between measuring voltage and measuring current.

Explain why the simulation must include numerical readings.

Explain how a reversed LED fault can be identified.

A strong short answer should not be vague. It should include the electrical reason, the expected result, and the correction when relevant.

6.9 Calculation Practice Questions

The calculation practice questions test whether the reader can perform the required numerical work independently.

A red LED has a forward voltage of 2V and is connected to a 9V battery. The desired current is 20mA. Calculate the required resistor.

A red LED has a forward voltage of 2V and is connected to a 9V battery through a 330Ω resistor. Calculate the current.

A red LED has a forward voltage of 2V and is connected to a 9V battery through a 360Ω resistor. Calculate the current.

A red LED has a forward voltage of 2V and is connected to a 9V battery through a 390Ω resistor. Calculate the current.

A red LED has a forward voltage of 2V and is connected to a 9V battery through a $10k\Omega$ resistor. Calculate the current.

If the resistor voltage is 7V and the current is 20mA, calculate the resistor power.

If the resistor value is 470Ω , calculate the LED current using the same 9V battery and 2V LED assumption.

If the desired current is 15mA instead of 20mA, calculate the required resistor.

If the LED forward voltage is 2.2V and the desired current is 20mA, calculate the required resistor.

If the supply voltage is reduced to 5V while the LED forward voltage remains 2V and the desired current is 20mA, calculate the required resistor.

The expected method for every resistor calculation is:

$$R = \frac{V_S - V_F}{I_F}$$

The expected method for every current calculation is:

$$I = \frac{V_S - V_F}{R}$$

The expected method for resistor power is:

$$P_R = V_R \times I$$

or:

$$P_R = I^2 R$$

6.10 Circuit Interpretation Questions

The circuit interpretation questions assess whether the reader can understand the schematic as an electrical structure rather than only as a drawing.

Identify the supply component in the circuit.

Identify the current limiting component in the circuit.

Identify the light emitting component in the circuit.

Identify the positive node of the circuit.

Identify the node between the resistor and LED.

Identify the negative return node of the circuit.

Trace the current path from battery positive to battery negative.

Explain why the resistor and LED are in series.

Explain why the same current flows through the resistor and LED.

Explain what would happen if the resistor were replaced by a wire.

Explain what would happen if the LED were removed.

Explain what would happen if the battery were removed.

Explain what would happen if the circuit loop were incomplete.

Explain how the LED polarity is shown in the schematic.

Explain how the resistor value affects LED brightness.

The correct interpretation is that the circuit contains one main current path. The resistor and LED are not independent branches. They are part of the same series loop. Therefore, changing the resistor value directly changes the current through the LED.

6.11 Proteus Practical Questions

The Proteus practical questions assess whether the reader can reproduce and verify the project inside the simulation software.

- Which Proteus workspace is used to draw the schematic?
- Which component is used to represent the battery?
- Which component is used to represent the resistor?
- Which component is used to represent the red LED?
- How is the battery voltage set to 9V?
- How is the resistor value changed?
- How is the LED connected in the correct polarity?
- How is the ammeter added to the circuit?
- How is the voltmeter added to the circuit?
- How is the simulation started?
- What should be observed when the circuit is correct?
- What measurement should be recorded from the ammeter?
- What measurement should be recorded from the voltmeter?
- How can the resistor value be changed to test troubleshooting cases?
- How should screenshots be captured for the report?

In the Proteus implementation, the report should include separate images for the final scheme, the running simulation, and the troubleshooting case. The scheme image should show correct wiring. The simulation image should show the LED operating with meter readings. The troubleshooting image should show a fault condition such as a 10k Ω resistor producing very low current.

6.12 Fault Finding Questions

The fault finding questions test whether the reader can diagnose incorrect circuit behavior from symptoms.

- The LED does not glow. What should be checked first?
- The LED is very dim. What resistor value problem may exist?
- The LED is too bright. What resistor value problem may exist?
- The ammeter shows zero current. What fault may exist?
- The voltmeter shows no voltage across the LED. What fault may exist?
- The circuit is connected but the LED is reversed. What result is expected?
- The resistor is missing. What risk exists?
- The battery voltage is not set to 9V. How will the simulation result change?
- The ammeter is connected in parallel by mistake. Why is this incorrect?
- The voltmeter is connected in series by mistake. Why is this incorrect?
- A wire is disconnected. What happens to the current?

The resistor value is accidentally set to 10kΩ. What happens to LED brightness?

The resistor value is accidentally set to 220Ω. What happens to LED current?

The LED model is changed to a different color. Why might the current change?

The simulation result does not match the calculation exactly. What should be checked?

A correct fault finding answer should include three parts: the likely cause, the electrical explanation, and the correction. For example, if the LED is very dim, the likely cause may be excessive resistance. The electrical explanation is that a high resistor value reduces current. The correction is to use the calculated resistor value or a suitable standard value near it.

6.13 Model Answers

This section provides model answers for the major question categories. The answers are written to show the expected academic standard of explanation. A correct answer should be accurate, complete, and directly connected to the circuit design.

6.13.1 Answers to Description Questions

The main purpose of the circuit is to connect a red LED safely to a 9V battery by placing a resistor in series with the LED. The resistor limits the current so that the LED can operate without being damaged by excessive current.

An LED should not be connected directly to a 9V battery because an LED does not behave like a simple current limiting device. Once the LED is forward biased, it allows current to pass, and without a resistor the current may rise to a damaging level.

The resistor is used as a protective component. It absorbs the remaining voltage after the LED forward voltage and converts part of the electrical energy into heat. By doing this, it controls the current flowing through the LED.

The circuit is called a current limiting circuit because the resistor determines the maximum current that can flow through the LED under the stated supply voltage and LED forward voltage assumptions.

The main components are the 9V battery, the resistor, the red LED, the connecting wires, and the measuring instruments used in Proteus.

Proteus is used because it allows the circuit to be drawn, simulated, measured, and tested before physical construction. It also allows fault conditions to be demonstrated safely.

6.13.2 Answers to Calculation Questions

Ohm's law states that voltage is equal to current multiplied by resistance:

$$V = IR$$

The resistor value can therefore be found by rearranging the equation:

$$R = \frac{V}{I}$$

For an LED circuit, the resistor voltage is not the full supply voltage. The LED uses part of the supply voltage as its forward voltage. Therefore, the resistor voltage is:

$$V_R = V_S - V_F$$

For this circuit:

$$V_R = 9V - 2V$$

$$V_R = 7V$$

The desired current is:

$$I_F = 20mA = 0.02A$$

Therefore, the required resistor is:

$$R = \frac{7V}{0.02A}$$

$$R = 350\Omega$$

A standard resistor value near this result should be selected. A 360Ω resistor gives approximately 19.4mA, which is very close to the 20mA target. A 390Ω resistor gives approximately 17.9mA, which is slightly lower and safer.

For 220Ω :

$$I = \frac{7V}{220\Omega}$$

$$I = 31.8mA$$

For 330Ω :

$$I = \frac{7V}{330\Omega}$$

$$I = 21.2mA$$

For 360Ω :

$$I = \frac{7V}{360\Omega}$$

$$I = 19.4mA$$

For 390Ω :

$$I = \frac{7V}{390\Omega}$$

$$I = 17.9mA$$

For $10k\Omega$:

$$I = \frac{7V}{10000\Omega}$$

$$I = 0.7mA$$

The resistor power at 20mA is:

$$P_R = V_R \times I$$

$$P_R = 7V \times 0.02A$$

$$P_R = 0.14W$$

The resistor should therefore have a power rating greater than this calculated value. A 0.25W resistor may be acceptable, while a 0.5W resistor gives a stronger safety margin.

6.13.3 Answers to Scheme Questions

The battery positive terminal should be connected to the first terminal of the resistor. The second terminal of the resistor should be connected to the anode of the LED. The cathode of the LED should be connected back to the negative terminal of the battery.

The LED anode must face the positive side of the supply, and the LED cathode must face the negative side. This arrangement forward biases the LED and allows current to flow through it.

The resistor must be in series with the LED because the same current must pass through both components. If the resistor is not in the LED current path, it cannot control the LED current.

The resistor may be placed before or after the LED in the same series loop. The current limiting effect is the same because the series current is identical at every point in the loop. However, placing the resistor between the battery positive terminal and the LED anode is clearer for educational presentation.

The ammeter must be connected in series because current must pass through the meter to be measured. The voltmeter must be connected in parallel because voltage is

measured across two points.

The correct current path is:

Battery positive → *Resistor* → *LED anode* →
LED cathode → *Battery negative*

If the LED is reversed, the LED is not forward biased and it will not glow under normal conditions.

6.13.4 Answers to Simulation Questions

The purpose of simulation is to verify the theoretical design before building the physical circuit. Simulation allows the student to check whether the LED receives the expected current and whether the voltage distribution matches the calculation.

The battery should be set to 9V. The resistor should be set to the selected design value, preferably 360Ω or 390Ω. The LED should be a red LED model. The ammeter should be connected in series, and the voltmeter should be connected across the LED.

With a 360Ω resistor, the expected current is approximately 19.4mA. With a 390Ω resistor, the expected current is approximately 17.9mA. The LED voltage should be approximately 2V, and the resistor voltage should be approximately 7V.

The LED glow alone is not enough evidence because brightness is a visual effect and may not show the exact current value. A correct simulation must include numerical measurement. The ammeter proves the current, and the voltmeter proves the voltage distribution.

If the simulation result differs from the calculation, the student should check the LED forward voltage model, resistor value, battery voltage, meter placement, LED polarity, and circuit wiring.

6.13.5 Answers to Troubleshooting Questions

If the LED does not glow, the first checks should be battery voltage, LED polarity, resistor value, wire continuity, and simulation setup.

If a 10kΩ resistor is used, the LED becomes very dim or invisible because the current is only about 0.7mA. This current is much lower than the normal operating current for a visible LED demonstration.

If a 220Ω resistor is used, the current is approximately 31.8mA under the stated assumptions. This is higher than the 20mA design target, so it is not the strongest safe value for the report.

If the LED is connected without a resistor, the circuit lacks a current limiting component. This creates a risk of excessive current and LED damage.

If the LED is reversed, it will not glow because it is not forward biased. The correction is to reverse the LED so that the anode faces the positive side and the cathode faces the negative side.

If the circuit has an open wire, the loop is incomplete and the current becomes

zero. The correction is to reconnect the missing wire and confirm a complete path from battery positive to battery negative.

If the ammeter is placed incorrectly, the current reading may be wrong or the circuit behavior may be disturbed. The ammeter must be placed in series. If the voltmeter is placed incorrectly, the measured voltage may not represent the LED voltage. The voltmeter must be placed in parallel with the LED.

6.14 Final Student Evaluation Checklist

The final student evaluation checklist confirms whether the student has completed the project with full technical understanding.

The student should be able to explain the purpose of the circuit.

The student should be able to identify every component in the circuit.

The student should be able to explain why the LED needs a resistor.

The student should be able to state Ohm's law correctly.

The student should be able to calculate the voltage across the resistor.

The student should be able to calculate the required resistor value.

The student should be able to convert milliamperes into amperes.

The student should be able to calculate current for different resistor values.

The student should be able to calculate resistor power dissipation.

The student should be able to select a suitable standard resistor value.

The student should be able to explain why 360Ω or 390Ω is better than 220Ω for a strict 20mA design.

The student should be able to explain why $10k\Omega$ makes the LED very dim.

The student should be able to wire the circuit correctly in Proteus.

The student should be able to place the ammeter correctly.

The student should be able to place the voltmeter correctly.

The student should be able to run the Proteus simulation.

The student should be able to compare simulation results with calculation results.

The student should be able to identify reversed LED polarity.

The student should be able to identify an open circuit.

The student should be able to identify excessive resistance.

The student should be able to identify insufficient resistance.

The student should be able to correct each fault using electrical reasoning.

The student should be able to prepare separate report images for the scheme, simulation, and troubleshooting sections.

The student should be able to answer conceptual, mathematical, practical, and diagnostic questions about the project.

6.15 Chapter Summary

This chapter evaluated the complete LED current limiting project through a structured set of questions and model answers. The questions were arranged according to the academic structure of the report, beginning with the project description and continuing through calculation, scheme design, simulation, troubleshooting, and practical Proteus implementation.

The chapter confirmed that the essential principle of the project is the safe control of LED current. A red LED connected to a 9V battery requires a series resistor because the LED cannot safely limit current by itself. The correct calculation subtracts the LED forward voltage from the supply voltage and then applies Ohm's law to the remaining voltage. With a 9V battery, a 2V red LED, and a 20mA design current, the calculated resistor value is 350Ω . A 360Ω resistor gives a current close to the target value, while a 390Ω resistor gives a slightly lower and safer current.

The chapter also confirmed that a correct schematic must show the battery, resistor, LED, measuring instruments, and correct polarity. The simulation must verify the circuit numerically by measuring current and voltage. Troubleshooting must explain why incorrect resistor values, reversed polarity, open wires, and incorrect meter placement produce wrong results.

The final conclusion of this chapter is that the student has mastered the project only when the circuit can be explained, calculated, connected, simulated, measured, diagnosed, and corrected without guessing.

